

Emerging Solid-to-Solid Phase-Change Materials for Thermal-Energy Harvesting, Storage, and Utilization

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Phase-change materials (PCMs) offer tremendous potential to store thermal energy during reversible phase transitions for state-of-the-art applications. The practicality of these materials is adversely restricted by volume expansion, phase segregation, and leakage problems associated with conventional solid-liquid PCMs. Solid-solid PCMs, as promising alternatives to solid-liquid PCMs, are gaining much attention toward practical thermal-energy storage (TES) owing to their inimitable advantages such as solid-state processing, negligible volume change during phase transition, no contamination, and long cyclic life. Herein, the aim is to provide a holistic analysis of solid-solid PCMs suitable for thermal-energy harvesting, storage, and utilization. The developing strategies of solid-solid PCMs are presented and then the structure-property relationship is discussed, followed by potential applications. Finally, an outlook discussion with momentous challenges and future directions is presented. Hopefully, this review will provide a guideline to the scientific community to develop high-performance solid-solid PCMs for advanced TES applications.

domestic and industrial sectors. To trade the issue of mismatch between energy supply and demand in terms of time and space, thermal energy is one of the most generous forms of energy to store renewable energy. Currently, 80–85% of the world's energy consumption involves the manipulation of thermal energy over a wide temperature range. Thermal-energy storage (TES) using latent heat of phase-change materials (PCMs) is one of the most adequate means of energy storage within small temperature constraints.^[8]

PCMs are an extensive class of materials, in which thermal energy is stored in the form of heat of fusion during the phase-change process and remains stored till the materials endure a reverse transition process.^[9,10] Phase transition temperature and heat of fusion are the requisite parameters of PCMs. The phase transition temperature is a temperature at which

1. Background

The disparate scarcity of fossil fuels and high-energy consumption of non-renewable energy have become serious concerns for the sustainability of energy and climate change. The high consumption of fossil fuels has directly led to carbon dioxide emissions that severely cause the energy crises and global warming.^[1–3] Therefore, researchers are firmly observing renewable energy sources such as solar, wind, hydro, and geothermal power because of their low carbon footprint and less air contaminant emission.^[4–6] Numerous technologies are being developed to harvest, store, and utilize renewable energy.^[7] However, the major drawback of these renewable sources is their intermittency, which limits their adoption in

material starts changing its states from one phase to another. The heat of fusion also termed as latent heat of fusion, refers to the amount of thermal energy absorbed or released during the phase transition of PCMs. The conceptual diagram of phase-change energy storage and release is presented in (Figure 1a). The working mechanism of PCMs involved storing and release of thermal energy at constant temperature followed by intermolecular interactions. The reversible breaking and formation of hydrogen bonding are accountable for the storage and release of thermal energy during melting and solidifying process. Under thermal charging, PCMs first perform like a sensible heat-storage material during which thermal energy is stored resulting in a temperature rise, but, when the melting temperature is achieved, then the materials are stored energy as latent heat from the melting process at a constant temperature. When melting is completed, further thermal action results in the rise of temperature analogous to sensible heat storage. Upon cooling of melted PCMs, the PCM starts to freeze and release an amount of latent heat about equal to melting enthalpy, when freezing temperature is attained. The thermo-physical properties of PCMs are governed by thermodynamics equilibrium,^[11] and can be represented by the Gibbs free energy equation;

$$\Delta G = \Delta H - T\Delta S \quad (1)$$

For a first-order phase transition, the Gibbs free energy of the solid phase equals to liquid phase and therefore $\Delta G_{S \rightarrow L} = 0$, giving the relation of $\Delta H_{\text{fusion}} = T_m \Delta S_{\text{fusion}}$. Where T_m indicates

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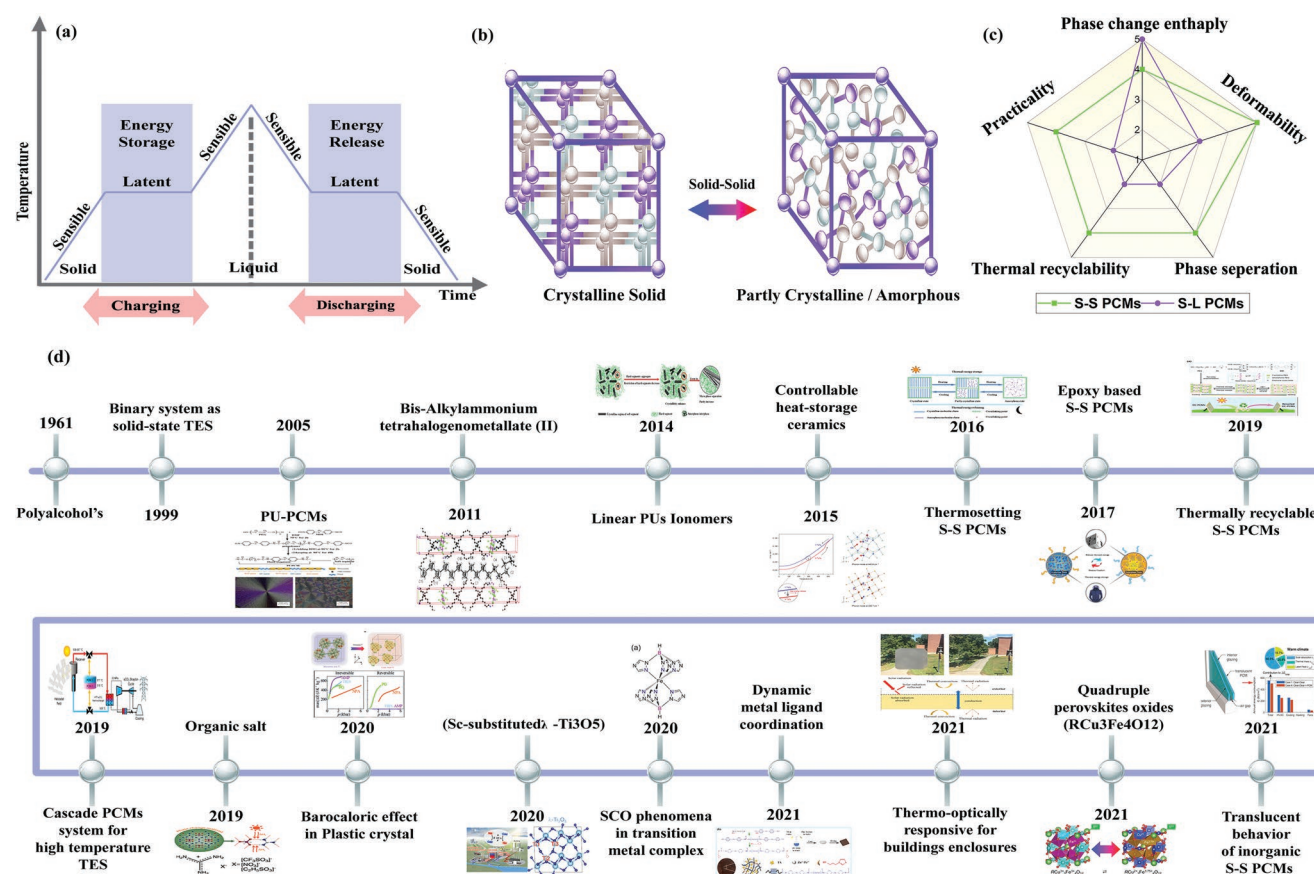


Figure 1. a) An illustration of the phase-change process. b) The working phenomena of S–S phase-change transition. c) The comparative characteristics of S–S PCMs and S–L PCMs. d) Representative highlights along with the major developments of S–S PCMs including polyalcohols, polymeric, inorganic, and hybrid-based materials for advanced TES. Image for “PU-PCMs”: Reproduced with permission.^[36] Copyright 2005, Elsevier. Image for “Bis-Alkylammonium tetrahalogenometallate”: Reproduced with permission.^[180] Copyright 2013, Elsevier. Image for “Linear PU Ionomers”: Reproduced with permission.^[40] Copyright 2014, Elsevier. Image for “Controllable heat-storage ceramics”: Reproduced under the terms of the CC-BY Creative Commons Attribution 4.0 International license (<https://creativecommons.org/licenses/by/4.0/>).^[149] Copyright 2015, The Authors, published by Springer Nature. Image for “Thermosetting S–S PCMs”: Reproduced with permission.^[39] Copyright 2016, Elsevier. Image for “Epoxy-based S–S PCMs” Reproduced with permission.^[40] Copyright 2017, Royal Society of Chemistry. Image for “Thermally recyclable S–S PCMs”: Reproduced with permission.^[90] Copyright 2019, Royal Society of Chemistry. Image for “Cascade PCMs system for high temperature TES”: Reproduced with permission.^[51] Copyright 2019, Elsevier. Image for “Organic salt”: Reproduced with permission.^[142] Copyright 2019, American Chemical Society. Image for “Barocaloric effect in plastic crystal”: Reproduced under the terms of the CC-BY Creative Commons Attribution 4.0 International license (<https://creativecommons.org/licenses/by/4.0/>).^[135] Copyright 2020, The Authors, published by Springer Nature. Image for “Sc-substituted λ -Ti₃O₅”: Reproduced under the terms of the CC-BY Creative Commons Attribution 4.0 International license (<https://creativecommons.org/licenses/by/4.0/>).^[152] Copyright 2020, The Authors, published by AAAS. Image for “SCO phenomena in transition metal complex”: Reproduced with permission.^[172] Copyright 2020, Wiley-VCH. Image for “Dynamic metal–ligand coordination”: Reproduced with permission.^[110] Copyright 2021, Elsevier. Image for “Thermo-optically responsive for buildings enclosures”: Reproduced with permission.^[198] Copyright 2021, Elsevier. Image for “Quadruple perovskites oxides (RCu₃Fe₄O₁₂)”: Reproduced with permission.^[181] Copyright 2021, Royal Society of Chemistry. Image for “Translucent behavior of inorganic S–S PCMs”: Reproduced with permission.^[199] Copyright 2021, Elsevier.

the melting temperature, ΔH_{fusion} is the measure of latent heat of fusion, and (ΔS), is the change in entropy after phase transition.

Generally, PCMs are classified into three main categories organic (paraffin, fatty acid, and polymers), inorganic PCMs (salt hydrate, metal, and alloys), and their eutectic mixtures based on their composition.^[9] They typically undergo four kinds of phase transition processes solid-to-liquid (S–L), solid-to-solid (S–S), solid-to-gas (S–G), and liquid-to-gas (L–G) phase transition, or vice versa. Among them, S–L and S–S phase transitions are the most suitable for TES-related applications.^[12–14] The S–L PCMs change their states between solid and liquid

on account of the reversible breaking and formation of inter-molecular hydrogen bonding or supramolecular bonding upon thermal action, which are accountable for the storage or release of thermal energy correspondingly. These PCMs have been broadly covered by various researchers in the past for the applications of TES including solar-TES, off-peak electricity, and infrared stealth technologies.^[9,15–24] Nevertheless, S–L PCMs suffer from high volume expansion and leakage of the liquid PCM during multiple phase-change cycles, which hindered their utilization at advanced levels.

In this regard, tremendous attention has been paid to encapsulate S–L PCMs to prevent leakage and make them

reliable for practical TES. However, this technology is still limited to producing high-performance encapsulated PCMs on large scale. For instance, nano-encapsulated PCMs exhibit better performance and shape-stability but mass production of these PCMs is challenging. On the other hand, macro-encapsulated PCMs suffer from poor thermal properties and large volume expansion which ultimately results in leakage over repeated cycles. Moreover, the inclusion of inactive shell materials or supporting materials causes a significant decline in energy storage capacity. In contrast to S–L PCMs, S–S PCMs go through less change in volume and entropy ensuring their effectiveness in practical TES. These PCMs offer an elegant solution to overcome these issues toward the potential of TES applications due to solid-state switching.^[25–28] S–S phase transitions are usually observed between the different crystalline forms of materials or polymorphs in a single component system. The working attitude of these materials involves molecular re-orientations, from one ordered crystalline phase to another disordered semi-crystalline solid phase without shifting into isotropic liquid phase to store or release thermal energy by changing temperature or pressure (Figure 1b). They generally exhibit first-order phase transitions that undergo discontinuous changes in phase-change enthalpy, changes in volume, and entropy owing to a crystal-packed structure. The extent of these changes is often observed much less as compared to conventional S–L PCMs, which makes them preferable and optimum choice in the field of TES applications. Further, these PCMs also present desirable characteristics such as smaller phase separation and no degradation upon thermal cycling.^[29,30] Even though the mechanical strength slightly shrinks after phase transformation in some cases of S–S PCMs, they normally maintain their bulk solid properties in a certain range of temperature, and their core role regarding leakage remains intact, thus they are also known as solid-state PCMs. The inherent thermal characteristics of S–S PCMs over S–L PCMs are presented in (Figure 1c).

Despite the growing importance of S–S PCMs, in-depth analysis of these PCMs is not sensed about the fundamentals of structural–thermal relationship with mechanical and physical properties toward TES. Therefore, there is an urgency to talk about the latest scientific literature progress on S–S PCMs in a thought-provoking way. Significant advancements have been made in the past on S–S phase transition materials for thermal-energy applications by means of different materials polyalcohols, polymeric, inorganic, and hybrid which are briefly summarized in (Figure 1d). To the best of our knowledge, only one single report has been made on S–S PCMs. However, they only focus on one type of molecular study at a smaller scale and did not study thermal-energy systems toward the adoption of these materials at a broader scale.^[28]

Here, we provide an inside view of various types of S–S PCMs, from the point of molecular engineering, solid-state phase transition, structure–plasticity property correlation, and topological arrangement identifying the execution of these materials into an unexplored sector of the field. We hope that this review will support the scientific fraternity to view a comprehensive understanding of materials design developments and their innovative results in the field of S–S PCMs for high-tech TES applications.

2. Overview of S–S PCMs

S–S PCMs are divided into three major families of materials to study their molecular and structural rearrangement occurring in solid-state phase transitions for the potential of TES has been illustrated in **Figure 2**. Organic molecular crystals exhibit energetic S–S phase transitions from one crystalline structure to another crystalline lattice configuration with the ease of polymorphic behavior. The family of organic S–S PCMs mainly includes polyalcohols/plastic crystals (PCs), polymeric, and organic-salt-based materials. Generally, organic S–S PCMs lie between 25 and 185 °C with their phase-change enthalpies from 15 to 270 kJ kg^{−1}. Polymeric S–S PCMs involve reversible phase transitions from a crystalline solid phase to another semi-crystalline or amorphous phase. Poly(ethylene glycol) (PEG) and poly(ethylene oxide) (PEO) are primarily employed as phase-change units incorporated into polymer skeleton through different chemical cross-linking polymerizations methods for the assembly of polymeric S–S PCMs. These materials are effectively corresponding to the low- and mid-range of temperatures that lie between 15 and 68 °C with phase-change enthalpies of 15–183 J g^{−1}. Polyalcohols is a sub-class of organic PCMs that undergo solid-state transformations from layered or chain structures to partially disordered ones with suspected rupture of hydrogen bonding over the function of temperature or pressure from low to high. These materials are typically applied in thermal-energy applications, that is, solar water heating and domestic heating owing to their reversible crystallographic arrangement. PCs also exhibit solid-state phase transformations analogous to polyalcohol. Recently these materials have gotten substantial attention as an alternative eco-friendly solution for refrigeration systems due to their magnificent behavior toward stimuli-controlled phase transition within solid-state processing.

Organic salt/protic ionic liquids are also another sub-class of organic PCMs that usually undergo S–L and S–S phase transition. These PCMs normally operate in the temperature range of 90–208 °C during S–L transitions for solar TES and wind applications. But, herein we only focus on such salts that exhibit S–S phase transitions. For instance, pyrazolium sulfonate [Pzy][CF₃SO₃], guanidinium triflate [Gdm][CF₃SO₃], and guanidinium ethane sulfonate [Gdm][C₂H₅SO₃] are the only salts that are presenting S–S phase transition in the temperature range of 59–115 °C for domestic water-heating applications. They mainly consist of organic cation and organic/inorganic anions or the substituent group and hydrogen bonding is the main driving force within the crystal packing structure during transitions. Inorganic S–S PCMs are such kinds of PCMs that can reversibly store and release thermal energy through order–disorder transition with the help of crystallographic structure transformation and magnetic transformation. The inorganic PCMs family mainly includes metallic and ceramics-based materials. The phase transition temperatures of these materials range from 30 to 120 °C with phase-change enthalpies falling between 25 and 70 J g^{−1}, which is relatively less than organic S–S PCMs. Further, inorganic-salt-based S–S PCMs are more suitable for high-temperature applications having an operating temperature range of 590–800 °C. Hybrid PCMs are another kind of S–S PCMs that is a combination of organic and inorganic materials

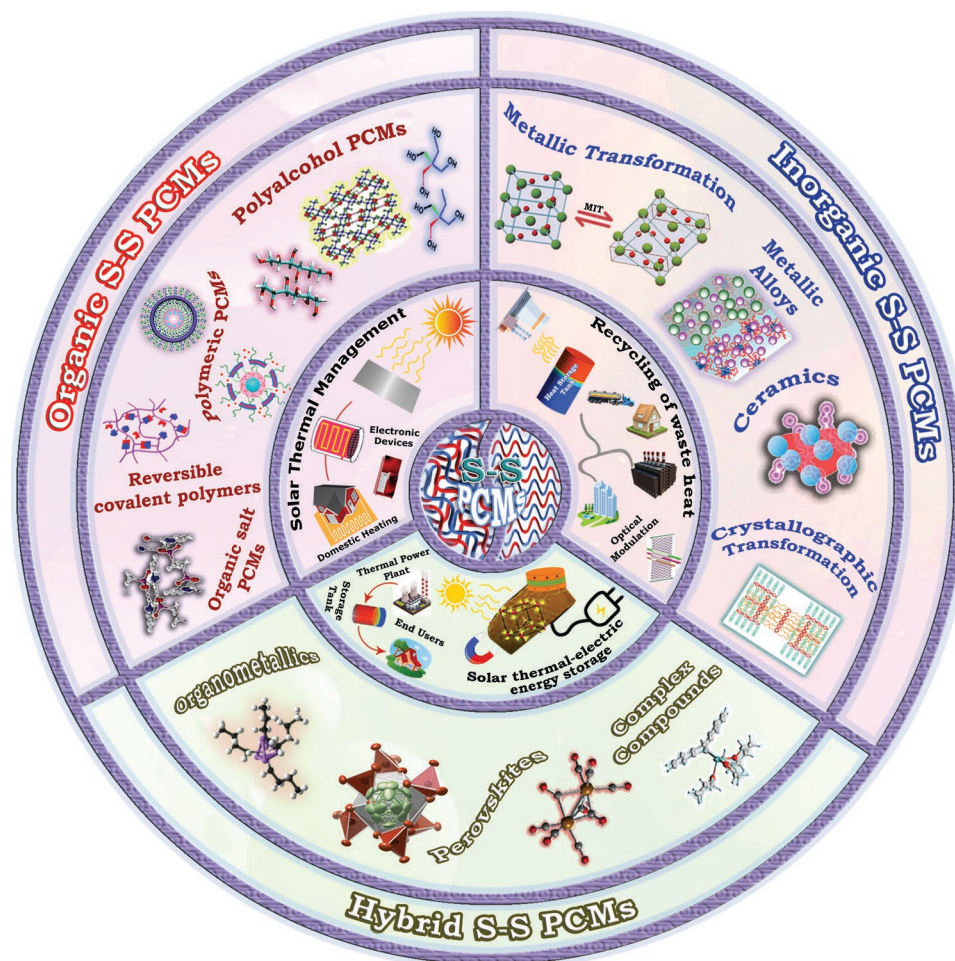


Figure 2. Structure of the review. Highlighting the major family of S–S PCMs with structural orientation and their prospective applications.

mainly including perovskites and organometallic-based materials. Hence, this article is categorized into three major categories of S–S PCMs including organic, inorganic, and hybrid materials to discuss their molecular structure and topological arrangement for advanced TES applications. The widespread adoption and commercialization of S–S PCMs are directly associated to a notable extent with their abundant availability and cost. Most of the organic S–S PCMs have been prepared with extensively available, low-cost starting materials which makes a straightforward pathway toward practicality. In contrast, inorganic S–S PCMs are typically obtained by metallic alloys, and optimization is needed to tune the intrinsic properties relationship when integrated for advanced utilization. The hybrid S–S PCMs have been prepared by relatively complex synthetic schemes that ultimately increase mass production costs. So, in the present situation, emerging categories of organic S–S PCMs including polymeric and PCs are being considered an optimal choice for innovative applications likely wearable thermal management applications and solid-state refrigeration technology applications, respectively. The phase transitions temperature of these materials is near ambient temperature, which is a necessary parameter when considering these applications. In addition, the flexibility and thermal storage capacity of these mate-

rials can be tuned by controlling the molecular weight for wearable applications.^[31,32]

On a concluding note, certain parameters are required regarding materials consideration and commercial-engineering aspects toward potential thermal-energy applications. For instance, thermal management of the human body by using S–S PCMs has been extensively studied for the fabrication of smart textiles to solve the leakage issue of liquid PCMs during phase transition, which is the main hindrance in PCM-integrated textiles. The key problem is significantly addressed by using polymeric-based S–S PCMs by providing a relatively stable and comfortable temperature range for human. Further, these materials also possess adjustable latent heat, negligible volume expansion, and decent mechanical properties rather than conventional S–L PCMs. For solid-state refrigeration technology, polyalcohols-based S–S PCMs normally exhibit a larger change in isothermal entropy and adiabatic temperature during the solid-state transition from ordered to disordered phase, making them a suitable choice in comparison to S–L PCMs. Although, these PCMs exhibit relatively better entropy change and latent heat storage. However, the large working temperature of these PCMs is challenging. On the other hand, the reuse of waste heat from an industrial power plant requires controlled

heat-release PCMs. These materials stored latent heat for longer and later, released on demand followed by external stimuli, regardless of temperature confines as observed in normal S–L PCMs. Detailed discussion of various S–S PCMs toward their potential thermal-energy applications is found in the corresponding Sections 3.1–3.3, respectively.

3. Strategies for Developing S–S PCMs

Until now, several developing strategies have been proposed for the functioning of S–S PCMs. The developing strategies mainly include molecular rearrangement likely face center lattice to cubic crystal lattice or crystalline region to gel formation state-amorphous region for polyalcohol and polymeric S–S PCMs, respectively, suggesting that the conversion of these materials purely occurs in solid state. These PCMs are typically developed by two kinds of strategies that include certain phase transition mechanisms for different materials. In the first strategy of phase transitions, a specified type of S–S PCMs undergoes an S–S phase transition from one crystalline phase to another crystalline phase with the associated energy fluctuation to absorb or release heat in terms of molecular arrangement. For instance, polyalcohols exhibit an S–S phase transition from sheet-like

crystalline structures to face-centered-cubic (FCC) or body-centered-cubic (BCC) crystal structures. The basic phenomena of these materials involve reversible phase transition from an ordered and layered structure like tetragonal, monoclinic, or orthorhombic crystal geometry to orientationally disordered phases such as FCC or BCC crystal with great symmetry orientation (Figure 3a). In short, these molecules involve phase transformation from homogeneous phase to heterogeneous phase upon external effect, that is, temperature or pressure difference while enduring a first-order transition. Therefore, these kinds of materials are also known as PCs representing the solid-state operation in their crystal structure.^[33–35]

While, in the case of polymeric S–S PCMs, phase-change motifs are typically integrated into a secondary structure via chemical covalent bonding. This phenomenon helps to prevent leaking the crystallizable constituents when they are in a liquid state (non-crystalline state). Because, the phase-changing constituents are structurally integrated into the macromolecular backbone of polymers through block polymerization^[36] or cross-linking polymerization with non-crystallizable moieties,^[37–42] side-chain grafting,^[43–45] or hyper branching.^[46,47] For instance, PEG or PEO are mostly used as the soft segment to attach with polymer skeleton as a hard segment for making linear chains of polymeric S–S PCMs due to the significant advantages of

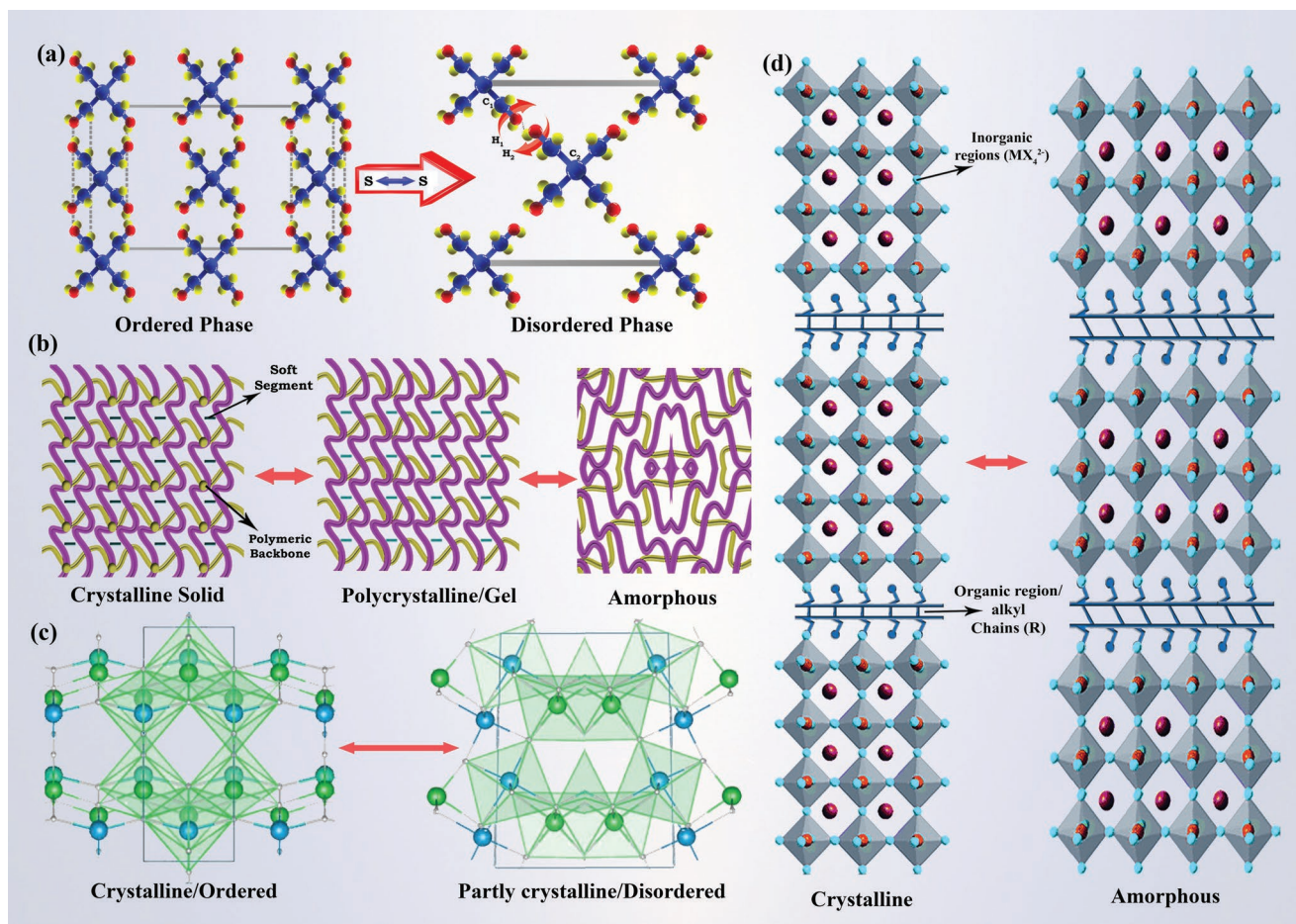


Figure 3. Developing strategies and phase transition mechanisms for different kinds of S–S PCMs. a) Polyalcohol-based S–S PCMs. b) Polymeric-based S–S PCMs. c) Inorganic S–S PCMs. d) Hybrid S–S PCMs.

PEG: good biocompatibility, excellent biodegradability, high latent heat, and operational simplicity.^[48,49] The PCMs absorb heat through latent heat mechanisms when the soft segment approaches phase transition temperature. The free movement of these soft segments is hindered, as they develop cross-linking network by the backbone of the polymer skeleton that allows the system to keep it in a solid state. The transition temperature of these S–S PCMs could be tuned by varying the chain length of the polymer backbone, adjusting the ratio of soft and hard segments, or changing the soft segments^[50] (Figure 3b). A detailed discussion about thermo-physical properties of polyalcohols and polymeric PCMs is provided in latter Section 3.1.

Inorganic S–S PCMs usually involves the absorption and release process of thermal energy upon magnetically induced transition, order–disorder, or crystalline phase transformation, similar to polyalcohols phase transition methodology (Figure 3c). For instance, the development of phase change within ferrite (Fe) and austenite (As), as well as inorganic salt like lithium sulfate, involves a solid-state phase transition.^[51,52] Examples of inorganic S–S PCMs include vanadium dioxide (VO₂), which typically undergo monoclinic to insulator phase transition over-temperature difference. By tuning the phase transition temperature or by developing the binary system with two different materials, such transformation can also be tailored for TES applications. A comprehensive explanation of inorganic S–S PCMs about thermo-physical properties and potential applications is provided in latter Section 3.2. Hybrid S–S PCMs are another class that includes perovskites and organometallic materials, in which inorganic 2D crystalline sheets as the supporting network are attached with crystallizing segments represented as “R” as alkyl chains through covalent bonding in the second methodology.^[53–56] Generally, hybrid S–S PCMs can be defined with the chemical formula $(nC_nH_{2n} + 1NH_3)_2MX_4$, where nC_nH_{2n} represents long alkyl chains with inorganic counterparts (MX_4^{-2}), M indicating a metal atom and X representing halogen group (Figure 3d). It generally exhibits analogous behavior with polymeric S–S PCMs in terms of their transitions. When these segments start absorbing heat energy, crystallizing segments start melting and storing thermal energy. In the reverse process, cooling from its melted state leads to heat release in a reversible process. Nonetheless, the lateral movements of molecules are hindered due to the attachment of inorganic materials (sheet-like scaffolds) with the soft segments. An inclusive discussion regarding thermal analysis and other properties of the hybrid PCMs is given in the Section 3.3.

The differences in thermal properties (phase transition temperature T_i and latent heat ΔH) of S–S PCMs mainly depend on their material types, the strength of the bonds, and structures. Thus, different kinds of S–S PCMs cover wide transition temperature ranges with different phase-change enthalpy for certain TES applications. Accordingly, suitable phase transition temperature and high latent heat storage capacity are the core requirements for each applications scenario to satisfy their adaptability. Based on the applications, some additional parameters also need to be optimized. For instance, the thermal management of the human body for wearable applications pays more attention to multifunction and good mechanical properties such as flexibility and elasticity. Similarly, good washability is required for smart textiles. On the other hand, the reuse of

waste heat from an industrial power plant requires stimuli-controlled heat-storage materials, which stored latent heat for longer and are later released on demand irrespective of temperature barrier. Further, the large-scale heat storage represented by solar heat storage needs a high heat-storage density for solar thermal-energy applications. Thermal management of electronic equipment and vehicles requires more focus on efficient heat dissipation and capacity.

3.1. Organic-Based S–S PCMs

Organic S–S PCMs mainly comprise a specific group of hydrocarbons-based materials that include polyalcohols, or PCs and polymers. Polyalcohols or small polyols are hydrocarbons that have BCC tetrahedral molecular structures near ambient temperature. These polyalcohols and some of their amine derivatives materials typically store and release thermal energy through reversible solid-state phase transformation when reasonably small molecules are engaged in the reversible breaking of supramolecular interactions, that is, hydrogen bonds in the molecular vibrations and lattice vibrations as a function of temperature.^[57] Thus, these materials are suitable for cold storage containers and packaging applications in the pharmaceutical and food industry, respectively. In the following sections, we will highlight recent developments regarding polyalcohols and polymeric S–S PCMs for thermal management utilization, space heating in detail, and so on.

3.1.1. Polymeric S–S PCMs

Among organic S–S PCMs, polymeric S–S PCMs are extensively studied owing to their exceptional properties including operational simplicity, reasonable latent heat storage capacity, and widely selectable phase transition temperature with excellent shape stability and adaptability.^[36,40,58,59] These PCMs are usually prepared by two methods including physical and chemical methods. The physical approach mainly involves the direct dispersion of PCMs molecules into a polymer framework as supporting skeletons with higher melting temperatures. In this case, PCMs can be developed by physical bonding of soft segment or S–L PCMs into polymer skeleton through melt blending, soaking, and physical adsorbing action technique to develop a stable polymeric matrix that helps to prevent leakage problems during phase transition. The obtained polymer can maintain its solid state during the S–L phase transition of the PCMs compound until the temperature of the supporting polymer is below its melting point. Thus, these polymeric PCMs are generally termed form-stable PCMs.^[14] However, the dispersion of PCMs into polymer supporting matrix may cause large phase segregation problems during multiple melting and freezing cycles, which ultimately limit their practical application.^[36,60,61] The chemically cross-linking approach is another method to develop polymeric S–S PCMs including blocking copolymerization, chemical grafting, and chemically cross-linked polymerization to overcome this issue. The cross-linked polymeric S–S PCMs mainly entailed two units; crystallizable identities/soft segments are chemically grafted to a backbone of a polymeric skeleton as explained earlier.

The working phenomena of cross-linking polymeric S–S PCMs generally involved the phase change of the soft segment into another semi-crystalline or amorphous phase rather than transforming into a liquid phase during the heating process. And the crystal structure exhibits reversed behavior when they are cooled below phase-changing temperature. Therefore, polymeric S–S PCMs have been enormously applied in various fashions for effective utilization of TES systems due to their favorable characteristics; negligible volume change, cost-effective, no liquid or gas generation during phase transition, and longer life span. Nonetheless, a slight reduction in latent heat was typically observed in these PCMs, because the mobility of crystallizing molecules is restricted through cross-linking rigid segments. The extent of the decrease in phase-change enthalpy is based on two things; a choice of polymer backbone and phase changing group.^[60] Therefore, several modifications in materials design approaches have been reported to improve the latent heat of PCMs. Considerable efforts have been made in the past via integrating PEG chains into several kinds of polymer skeleton materials through different techniques including polystyrene-graft-PEG copolymers, grafting 1-octadecanol (ODT) onto an amine cured epoxy cross-linked network, polyesters, and cross-linked polyurethanes (PUs) for the fabrication of polymeric S–S PCMs.^[39,45,61–68] Further, various kinds of cross-linking segments including tri-hydroxy surfactants, that is, Tween 80 and span 80, hexahydroxy compounds, that is, dipentaerythritol, pentaerythritol (PE), sorbitol, and inositol as well as β -cyclodextrin were also being used for the preparation of PUs-based S–S PCMs.^[68–72] Among them, PUs-based S–S PCMs systems have earned remarkable attention for effective utilization of TES systems in real-life applications owing to their fascinating thermo-physical characteristics; excellent thermal stability and good mechanical properties. The prominent preparation method of PUs-based S–S PCMs is chemically cross-linking polymerization by which PEG and its derivatives are chemically covalently bonded with supporting skeletons like hexamethylene diisocyanate (HDI) and diphenylmethane diisocyanate (MDI) and then cross-linked with various chain extender, that is, 1,4-butane diol,^[36] PE,^[37] and PVA^[73] through the formulation of urethane linkages (Figure 4a). The TES performance of PUs-based S–S PCMs is primarily influenced by several factors; 1) varying the structure of a molecule, 2) free movement of the soft segment or crystallizing group, and 3) cross-linking density present in the network. In view of the TES performance and their applicability, significant advancements have been made by our group in the development of PUs-based S–S PCMs by varying PEG molecular weight to adapt its intrinsic properties toward state-of-the-art TES applications.^[31,32,74,75] For instance, a kind of flexible PCMs is particularly favorable for human body thermal management to thermo-regulated textiles or practical wearable applications. In this context, our group has made notable progress in the development of solar-driven thermal-storage fabric with the impregnation of phosphorene encapsulated into PU-based S–S PCMs system to improve the washability, which was also coated on commercial cotton fiber to check its compatibility with the skin for practical applications. In designed assembly, phosphorene transforms solar energy into heat

which is stored inside the PCMs following solid-state processing (Figure 4b).^[74] The recognized fabric stored latent heat at about 51.3 J g^{-1} , which is mainly depending on how much PCM content was used in the fabric. The exceptional solar energy storage performance of the integrated PCM displays remarkable potential in solar-driven thermal therapy of the human body as well as other heating devices. Further, ultra-flexible and reliable S–S PCMs based on PUs were also developed by our group for TES applications by varying the molecular weight of PEG segments and the degree of cross-linking density within the polymeric network (Figure 4c).^[32] It was determined that fabricated PUs-based S–S PCMs with a molecular weight (PEG-3000) exhibited favorable flexibility and reasonable latent heat storage capacity that making them an optimal choice for applications of flexible PCMs. The resultant PCMs are potentially applied for wearable smart textiles and thermal temperature control system in electronic devices owing to their favorable flexibility and mechanical properties over a wide range of temperature.^[32,76] The fabricated PCMs exhibited adaptable phase-change enthalpy and mechanical properties by tuning the molecular weight of PEG, endowed better thermal management for the human body and wearable devices. Moreover, recently explored flexible PCMs thin film can be considered for wearable devices. The authors developed a stable and super tough type of S–S PCMs with π – π stacking through a block copolymerization.^[77] In the presented design, PEG, as PCM units were primarily attached to the MDI, served as polymer skeleton and biphenol through a two-step cross-linking reaction. The rigid groups containing benzene rings within PCMs afforded strong intermolecular forces, that is, hydrogen bonding and π – π stacking, resulting in PCMs displaying robust mechanical properties and shape stability through the solid-state operation (Figure 4d).^[77]

Another recent and important development was made to design intrinsically flexible phase-change film through a chemically cross-linked network reaction with the combination of PEG, melamine, and toluene diisocyanate as a cross-linker reagent.^[78] The as-prepared PCMs film displayed ultra-flexible characteristics and exhibits a suitable phase-change enthalpy by tuning the phase transition temperature from 5 to 50°C to tailor its functionality toward potential applications. This is mainly attributed due to the chemically cross-linked PEG chains into the entire polymer network, which implies the presence of S–S phase transitions characteristics. In addition, these films exhibited excellent energy storage efficiency of $\approx 93.5\%$ which can be achieved with applied current 1.5 A, corresponding to stable energy conversion efficiency during the heating and cooling process (Figure 4e,f).^[78] Further, the temperature control performance test revealed a controlled temperature for stitched lab coats and bending items over the phase transitions temperature range of 39.5 – 49.5°C (Figure 4g) demonstrating their wearability and temperature control stability as these factors are highly desirable for the human body and portable electronics, respectively, in an extreme environment. The IR images of the lab coat presented excellent shape stability and flexible thermal control devices during the heating and cooling process in both conditions (sewed and bending) from 21 to 89°C .

On the other hand, several thermoset polymeric S–S PCMs were prepared through solvent-free radical polymerization in

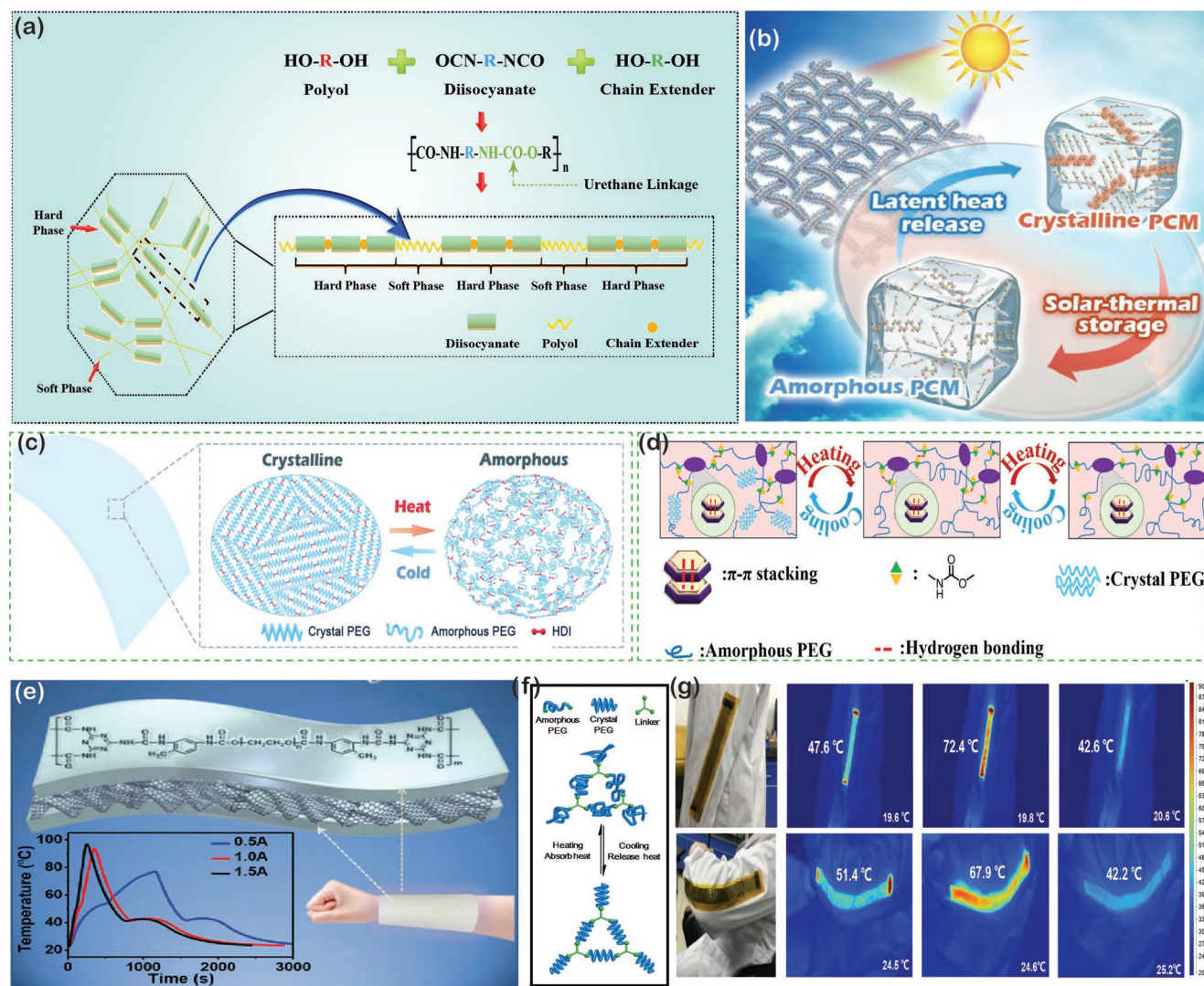


Figure 4. a) Schematic illustration of urethane linkages in terms of their soft segment and hard segment arrangement, which ensure the uniform dispersion of these segments within the PU network. b) Illustrative design of a PUs-doped solar-heat-storage fabric for thermal management of the human body. b) Adapted with permission.^[74] Copyright 2020, Elsevier. c) The schematic representation of cross-linked PU-based S-S PCM with its phase-change mechanism between crystalline and amorphous states. Adapted with permission.^[32] Copyright 2020, Royal Society of Chemistry. d) The working mechanism of PCMs during the heating and cooling process through π - π stacking. Adapted with permission.^[77] Copyright 2021, Elsevier. e) Schematic representation of intrinsically flexible phase-change film for wearable thermal managements with electrothermal energy efficiency. f) Systematic illustration of S-S phase transitions during the heating and cooling process. g) The execution of the materials on a sewed lab coat to view its shape stability and flexibility of thermal control device under electric field during heating and cooling process with a scale bar of IR images from 21 to 89 °C. e–g) Adapted with permission.^[78] Copyright 2021, Elsevier.

which PEG as a phase-change unit was permanently bonded to the polymer skeleton via the chemical cross-linking network.^[42,79,80] These S-S PCMs also endowed great potential for TES applications, that is, buildings and the textile industry due to the vigorous physical properties with good thermal stability and mechanical properties. Nevertheless, the majority of polymeric S-S PCMs were prepared by conventional chemical cross-linking methods to keep their shape stability above the phase switching temperature. Even though they have better physical and thermal properties with decent LHS capacity, they still have a big concern regarding workability and recyclability, as they cannot be reprocessed due to permanent or irreversible chemical bonds similar to thermosetting poly-

mers. Further, these PCMs generally do not show enough ductility due to the absence of solid-state plasticity during fabrication.^[39]

In this connection, influential strategies have been put forth through dynamic covalent chemistry including the formation of hydroxyl-ester bonds,^[81] hindered amine bonds,^[82] reversible dynamic urea bond,^[83] oxime carbamate bond,^[84] aliphatic disulfide bond,^[85] aromatic imine bond,^[86] and Diels–Alder reaction for the fabrication of recyclable polymers in the last few decades. These dynamic cross-link networks provide the possibility of obtaining a more reliable and thermally stable nature of covalent bonds than the conventional cross-linking networks. The dynamic covalent bond present in urethane,

ester, Diels–Alder, and oxime carbamate bond can be triggered by light, heat, and chemical reagents. Thus, polymeric S–S PCMs developed by the dynamic cross-linking network are being more realistic strategies and gaining significant interest for TES owing to their good solid-state plasticity, good mechanical property, and excellent recyclability, which ultimately help to limit the waste pollution.^[81,87–89] Wu et al. designed a novel type of S–S PCMs (DC-PCMs) by using a two-step dynamic cross-linking polymerization. In which the PEG segments serving as the PCM unit was first attached to the backbone of polymer scaffolds like HDI trimer and furfuryl alcohol and then the intermediate product was formed. In the second step, the intermediate product was further reacted with bis(4-maleinimidophenyl)methane through a thermally reversible Diels–Alder reaction connecting furan and maleimide (Figure 5a).^[90] PEG chain is mainly attributed to storing and releasing thermal

energy. The authors' finding recognized that the chemical cross-linking existing inside the PCMs is responsible for the formation of the S–S phase transition. The developed DC-PCMs structure presents good shape stability and keeps its original shape without any leakage behavior, as the PEG segments are chemically connected to the polymer backbone to retain their original shape. The polymeric framework in PCMs imparted shape stability and vigorous mechanical properties that can be modified via adjusting the PEG molecular weight. Further, thermally reversible Diels–Alder reaction with a relatively larger temperature (89.8–121 °C) than the phase switching temperature of PCMs varying between 39 and 71 °C exhibited the admirable thermal recyclability and solid-state plasticity of the designed PCMs. The S–S PCMs samples could be plastically redesigned into varied eternal forms at 120 °C because of the structural manipulations instead of displaying liquid

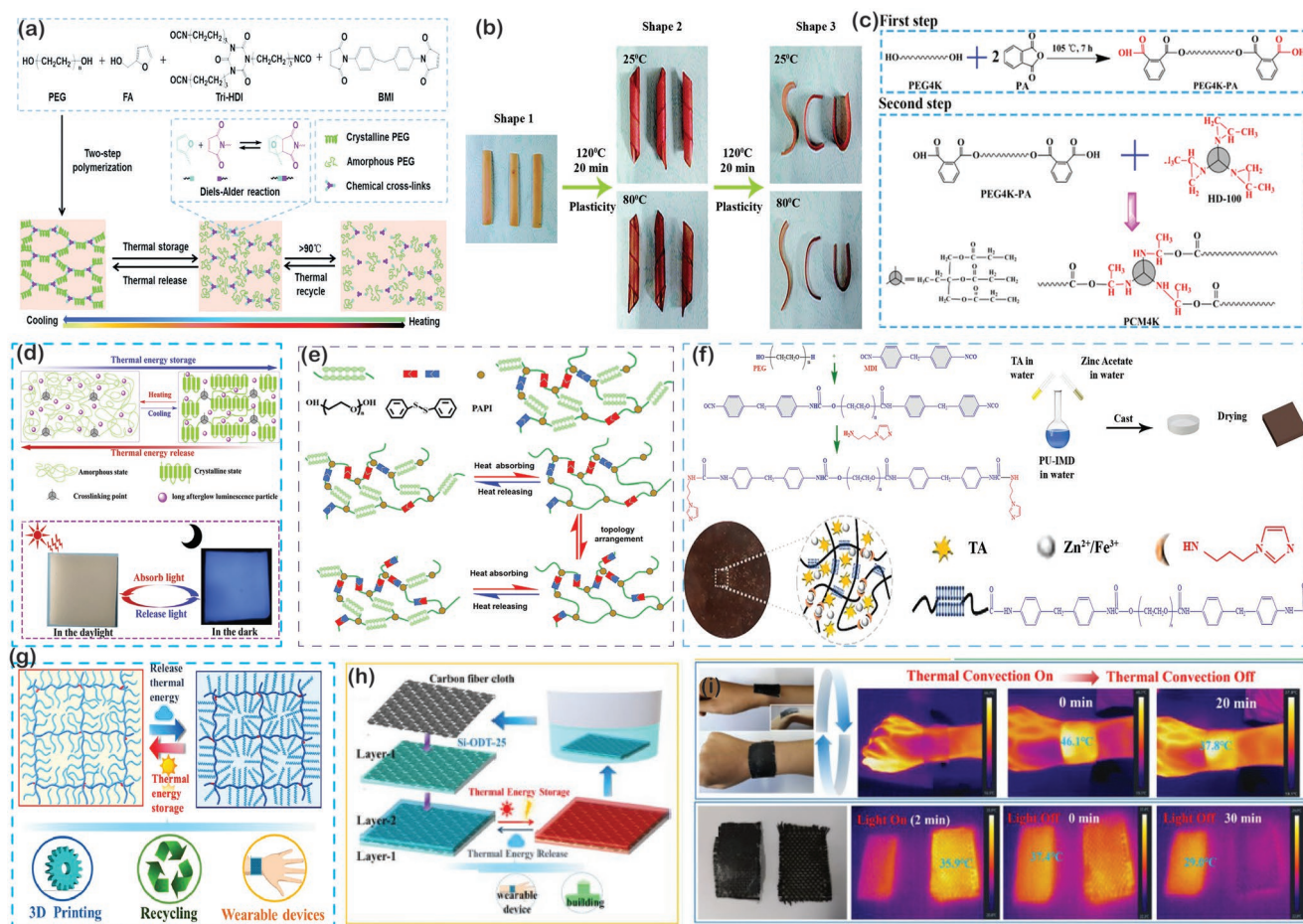


Figure 5. Development of thermosetting S–S PCMs through the dynamic covalent cross-linking network for thermal-energy storage. a) Schematic representation of the developed DC-PCMs by dynamic covalent cross-linking polymerization reaction through Diels–Alder reaction for reversible sustainable TES and thermal recycling. b) Solid-state plasticity of developed PCMs. Adapted with permission.^[90] Copyright 2019, Royal Society of Chemistry. c) Synthetic routes of self-luminous S–S PCMs. d) Illustration of the recommended mechanism for thermal and light energy storage for signboards. c,d) Adapted with permission.^[83] Copyright 2020, Elsevier. e) The chemical structure of raw materials and V-PCMs with a dynamic topology arrangement mechanism. Adapted with permission.^[89] Copyright 2021, Elsevier. f) Illustration of synthesis mechanism of the thermally stable S–S PCMs relying on dynamic metal–ligand coordination for well-organized sustainable TES. Adapted with permission.^[110] Copyright 2021, Elsevier. g) Schematic overview of the comb/bottlebrush polysiloxane networks (Si-ODT-x) via thiol–ene click chemistry with phase-transition behaviors. h) Illustrations of the fabrication process of the recyclable carbon fiber cloth by combing with phase-change polysiloxane composite for wearable devices. i) The wearable and thermal management properties of developed materials on wrist joint by thermal convection and solar–thermal energy conversion and storage performance. g–i) Adapted with permission.^[112] Copyright 2021, Elsevier.

variability (Figure 5b).^[90] Conclusively, the as-prepared DC-PCMs endowed myriad unique properties toward the potential of TES applications as compared to conventional cross-linked PCMs owing to the reversible energy storage ability excellent thermal reliability, and solid-state plasticity. Following similar routes, another report on PEG-containing S–S PCMs was successfully prepared through dynamic ferric ion cross-linking polymerization by Wang et al.^[91] The overall fabrication of developed S–S PCMs was a complex cross-linking network that involves four stages including preparation of pre-polymer, dissolved citric acid in DMF, formation of a carboxylate group (COO[−]) ended group pre-polymer by adding triethylamine owing to the interactions of COO[−] with ferric ion (Fe³⁺) leading to the formation of highly cross-linked network of S–S PCMs, followed by the cross-linking structure. The authors pointed out that, the developed polymeric S–S PCMs exhibit solid-state transition even at 95 °C, and no leaking of liquid was noticed during phase transition because of the dynamic ion cross-linking network structure. More importantly, it was concluded that the as-designed PCMs restrained their thermal characteristics like LHS capacity, crystallinity, thermal stability, and chemical structure even after recycling, which ultimately boost the thermal-energy performance for their widespread adoption in the various sector. Concerning their comparative characteristics, we highlight the key features of different polymeric S–S PCMs, and their thermo-physical properties have been explored (Table 1).

Lately, further development of additional functions has been carried out by the incorporation of afterglow-luminous (LAL) particles into PEG-based self-luminous S–S PCMs.^[83,111] In the developed structure, PEG chains were primarily bonded to the polymer skeleton of the phthalic anhydride (PA) via ring-opening reaction, and then the intermediate product was further reacted with a stoichiometric amount of trimethylolpropane-tris [3-(2-methylazopropyl)] propionate (HD-100) and weight percentage of LAL particles via two-step cross-linking reaction between trifunctional aziridine and intermediate product (Figure 5c).^[83] The polymeric skeleton and addition of LAL particles in S–S PCMs bestowed shape-stability, vigorous mechanical properties, and crystallinity that can be further modified by changing the molecular weight of PEG and the content of LAL particles. Further, the as-prepared PCMs can absorb/release a large amount of thermal energy reversibly from crystalline to amorphous states during the phase-change cycle. Owing to the addition of LAL particles, PCMs presented self-luminous ability via absorbing and storing visible light within PCMs in the daytime, which later on releases blue light gradually in the night for a long time making it an encouraging candidate for real-world applications, that is, emergency signs boards and wallboards buildings (Figure 5d).^[83] Very recently, significant advancements are being made in polymeric S–S PCMs to prepare recyclable thermosetting materials through dynamic covalent adaptable networks (CANs) to improve the service life of materials for excellent multifunctional TES applications. The dynamic bond can be activated by light, heat, and chemical reagents that directly led to the topology arrangement of CANs; impart the cross-linking polymers with recyclability, and remolding ability. Therefore, CANs are meaningfully considered to develop chemically cross-link S–S PCMs possessing

robust mechanical properties, solvent resistance, and recyclability over solid-state transition.^[105,107] In this regard, a novel dynamic covalent-cross-link PCMs termed (V-PCMs) were mainly developed by PEG as PCM component and polyaryl polymethylene isocyanate (PAPI) as cross-linking constituents, which help to improve the stability of V-PCMs and disulfide as dynamic bonds. It was found that the designed PCMs can retain their cross-linking network even above topology freezing transition temperature (T_v) around 135 °C proving the existence of an associative dynamic mechanism present inside the network structure. The developed PCMs can keep their reliability during the phase transition process on account of the 3D cross-link network, which further assists them to exhibit excellent anti-leakage performance even in the topology arrangement process (Figure 5e).^[89] V-PCMs can reversibly store and release heat in an ambient environment followed by the melting and crystallization cycle of PEG below T_v. As the temperature exceeds T_v, disulfide bond is being triggered resulting in a dynamic-exchange reaction happening, which led to the topology arrangement with recyclability and reprocessability. The exchange reaction mainly attributed to the dynamic CANs was done over the balanced mixture of disulfide bonds within the cross-linking network structure. In a similar direction, a new kind of S–S PCMs network was developed through supramolecular cross-linked interactions followed by the integration of dynamic metal–ligand bonds.^[110] In the synthesis mechanism, PEG was first bonded with the polymer chain of MDI and IMD-NH₂ to develop S–S PCMs (DS-PCMs), which were further reacted with metals salt of Zn and Fe via the supramolecular cross-link coordination bonding reaction between Zn²⁺/Fe³⁺ and tannic acid with imidazole group. The reversible TES and reprocessability process mainly include two stages of representation. The phase-change process first occurs below the phase transition temperature around 55–65 °C, as the PEG can store and release thermal energy reversibly in the DS-PCMs network. While in another stage, the dynamic supramolecular cross-link can be thermally recycled and reversibly reprocessed over reversible metal–ligand bonds in the temperature range of 80–120 °C (Figure 5f).^[110] The explored PCM can be adapted in a wide range of nearly 26–90% of a covalent bond employing remarkable self-healing and stimuli-responsive characteristics ability that has a significant impact on self-healing characteristics for elastomeric materials. In addition, the as-prepared PCMs endowed phenomenal recyclability and virtuous thermo-plasticity owing to the metal–ligand coordination chemistry. Thus, these materials are suitable for sustainable energy storage systems and thermal management applications.

Further recent and significant advancements have been made toward sustainable TES and wearable devices. In this design, a series of 3D printable and adjustable comb/bottlebrush phase-change polysiloxane networks (Si-ODT-*x*) were prepared by grafting or cross-linking alkyl chains ODT onto the poly(methylsiloxane)s with different HDT as cross-linkers over UV irradiation followed by thiol–ene reaction (Figure 5g).^[112] They further developed recyclable and multiresponsive PCM by combing phase-change polysiloxane composite with carbon fiber cloth (CFC) to view the adaptability of designed materials for wearable applications. The authors finding exhibited that flexible CFC displayed a good bonding

Table 1. The overall summary of polymeric-based S–S PCM for thermal-energy applications.

PCMs	Type of transition	Polymerization approach	T_m [°C]	ΔH_m [J g ^{−1}]	T_c [°C]	ΔH_c [J g ^{−1}]	Application	Reference
PUPCM	S–S	Blocked	65.28	138.7	38.5	126.2	TES	[36]
PEG-HBPU	S–S	Blocked	67	138	33	132	TES	[46]
PEG	S–L	–	54.4	174.5	43.8	162.1	TES	[32]
PU-PCMs	S–S	Blocked	57	142	33	137	Thermal control	[40]
HB-PUPCMs	S–S	H-branching	66	124	33.1	121.4	TES	[92]
HBPU	S–S	H-branching	55	115	30	112	TES	[47]
D-HBPU	S–S	H-branching	59	125	36	120	TES	[47]
Cellulose-g-PEG	S–S	Grafting	42–60	77–150	26–40	44–110	TES	[48]
Polystyrene-g-PA	S–S	Grafting	18–22	26–40	18.7	20–39	Floor heating	[43]
PA	S–L	–	61–63	203	–	–	TES	[14]
PVA-g-Cl8OH	S–S	Grafting	55.1	47	46	49	TES	[93]
Cl8OH	S–L	–	62	256.7	58	255.3	TES	[14]
PS-g-PEG ₆₀₀₀	S–S	Grafting	44–58	111–179	34.38	99–143	Space heating	[62]
PAN-g-MAPEG	S–S	Grafting	51.4	74.16	16.2	65.4	TES	[94]
SAN-g-PA	S–S	Grafting	29–37	6.9–10	31–41	12–24.4	TES	[95]
SAN-g-PEG	S–S	Grafting	49	68.2	30	66.8	TES	[44]
Cellulose-g-E ₂ C ₁₆	S–S	Grafting	28.6	33	14.9	30	TES	[96]
PAA-g-MPEG	S–S	Grafting	47–53	48–85	11–22	44–83	TES	[97]
MMA-g-PEGA	S–S	Grafting	53	160.5	43	158	TES	[98]
PS-Co-MA)-FA	S–S	Grafting	40–67	43–127	37–63	32–125	Thermal control	[29]
MA	S–L	–	51.5	193.7	–	–	TES	[14]
PSAA-g-PEG	S–S	Grafting	25–57	41–142	17–45	43–137	Thermoregulations	[60]
SA	S–L	–	69.9	222	–	–	TES	[14]
PEG/MDI/PE	S–S	Cross-linking	58.6	152.9	–	–	TES	[37]
PEG/MDI/PVA	S–S	Cross-linking	61.1	72.8	34	–	TES	[73]
MPEG/TDI/styrene	S–S	Cross-linking	57	108	19	81	TES	[64]
PGMA/PEG	S–S	Cross-linking	60	70	25	67.2	TES	[99]
TPU-PCMs	S–S	Cross-linking	56	137	37	127	TES	[72]
PU-s-PCMs	S–S	Cross-linking	48	153	23	143	TES	[68]
MF/PEG	S–S	Cross-linking	46–54	91–109	25–36	85–103	TES	[100]
PEG-sorbitol-PU	S–S	Cross-linking	59	107	45	102	TES	[70]
PEG-PUPCMs	S–S	Cross-linking	44–47	122–127	38.5	126	TES	[69]
DMPEG–PU	S–S	Cross-linking	59	121	29	118	TES	[50]
PEG/MDI/ β -CD	S–S	Cross-linking	58.9	110.5	46.3	103.5	TES	[71]
PEG/MDI/Castor	S–S	Cross-linking	51.4	117.7	42.3	109	Solar TES	[66]
Thermoset PU	S–S	Cross-linking	49.2	98.5	38.2	102.4	TES	[39]
HP-PU	S–S	Cross-linking	56.5	123.5	33.6	120.1	TES	[101]
ODT-g-DADGEBA	S–S	Cross-linking	36.5	14–70.5	24.7	14.3–65	Smart fibers	[65]
ODT	S–L	–	56	253.7	57.5	252.3	TES	[65]
PEG/PAPI	S–S	Cross-linking	50.4	111.7	41.3	110.4	TES	[80]
PEG/CA/TMPTP	S–S	Cross-linking	53	108	43	106	TES	[102]
Xylitol/PU	S–S	Cross-linking	38–41	64–76	24–29	68–80	TES	[42]
PEG/PMMA	S–S	Cross-linking	29.6	71.1	65.4	19.3	TES	[79]
PEG/HTPDMS	S–S	Cross-linking	62.9	124.5	25.7	104.4	TES	[103]
PEG/HDIT-PU	S–S	Cross-linking	65	136.8	34.1	–	Solar TES	[41]

Table 1. Continued.

PCMs	Type of transition	Polymerization approach	T_m [°C]	ΔH_m [J g ⁻¹]	T_c [°C]	ΔH_c [J g ⁻¹]	Application	Reference
DC-PCMs	S–S	D-cross-linking	71	87.8	39.2	107.2	Sustainable TES	[90]
PEG/PA/HD-100	S–S	Cross-linking	27–28	70–118	50–52	72–120	Emergency signs	[83]
PEG/HDI/CQD	S–S	Cross-linking	39.3	95.4	27.1	80.4	TES	[104]
PEG/IPDI/TEA	S–S	D-cross-linking	45.9	97.6	25.8	92.4	TES	[91]
OC-DCPs	S–S	D-cross-linking	55	101	34	99	Renewable TES	[105]
Azobenzene/ MR-PUss	S–S	Cross-linking	47	85–87	–	–	TES	[106]
PEG/MEM	S–S	Cross-linking	28	79	–	–	Sustainable TES	[59]
Polyester-PCMNs	S–S	D-cross-linking	51	120	60	–	TES	[107]
PCMNs	S–S	D-cross-linking	51	116	37	111	TES	[108]
PEG-PUss	S–S	D-cross-linking	49	85	39	87	TES	[109]
V-PCMs	S–S	D-cross-linking	43	105	32	101	TES	[89]
DS-PCMs	S–S	Metal–ligand	55	84–95	23	80–99	TES	[110]

ΔH_m and ΔH_c represent the melting and crystallizing latent heat, respectively; T_m and T_c signify the melting and crystallization temperatures, respectively. H-branching is for “hyper-branching” and D-cross-linking represents “dynamic cross-linking”

interface with phase-change polysiloxane network due to the intrinsic elongation of the material, after curing through UV radiation (Figure 5h).^[112] Consequently, layer one architecture of recyclable CFC exhibited fine adhesiveness toward the skin surface over certain pressure that helps the materials to avoid coming down during bending of the wrist. The adhesives characteristics on the wrist joint are mainly attributed due to the van der Waals interactions per unit area and comb-like network structure that ultimately led to robust mechanical inter-twines with rough surfaces at body temperature. The developed PCM on the wrist joint was heated to 45.1 °C followed by thermal convection and then drop down gradually to 36.8 °C after 20 min which signifies its importance to temperature management and heat therapy in the human body. Besides, it can also be suitable to collect thermal energy from sunlight and external source because of the photo-thermal and electric conductivity behaviors of carbon fibers with a phase-change polysiloxane network (Figure 5i).^[112] Furthermore, 3D printing technology was suggestively applied in the TES field for commercial liquid crystal desktop display followed by stereolithography.

Conclusively, thermal management of the human body by using S–S PCMs has been extensively studied in the fabrication of smart textiles to solve the leakage issue of liquid PCMs. For example, PUs-based PCMs are enormously used as heat storage in smart textile applications owing to their adjustable latent heat, negligible volume expansion, reversible nature of phase transitions, and decent mechanical properties. The S–S PCMs-based textiles are considered to be smart because they react to changes in environmental temperature by absorbing and releasing latent heat, providing a thermo-regulating effect. The excellent latest findings with key insight on polymeric S–S PCMs are significantly covered in this section, which opens up infinite new opportunities to fabricate PCM-containing smart textiles for thermal management of the human body in terms of wearable and thermotherapy-related applications. In summary, numerous statistics from the literature regarding

polymeric S–S PCMs for practical TES applications have been reviewed to understand the effect of the phase-changing motif and polymer macromolecular architecture on latent heatstorage capacity, see Table 1.

3.1.2. Polyalcohols

Polyalcohols, especially small polyols, are potential candidates for TES applications that undergo a solid-state phase transition. These molecules usually undergo reversible phase transitions from a low temperature ordered layered structure, that is, tetragonal, monoclinic to a high temperature orientationally disordered FCC or BCC phase, referred to as plastic or orientationally dispersed crystal. Thus, these materials are also known as PCs. Fundamentally, PCs are a sub-class of polyalcohols, however, all the polyalcohols and amine derivatives do not give the phenomena of PCs. These molecules usually undergo first-order solid-state transitions from low temperature ordered layered structure to an orientationally disordered phase known as the PCs phase.^[33,113]

Polyalcohols-based PCMs have been investigated first as solid-state phase transition since the late 1961s, by J. Timmermanns.^[33] In this direction, several papers regarding polyalcohol and its amine derivatives have been reported in terms of their thermodynamic and crystallographic properties for advanced TES utilization.^[35,114–121] More importantly, these highly symmetric polyalcohols mainly included PE, neopentylglycol (NPG), tris(hydroxymethyl)amino methane (TRIS), pentaglycerine (PG), trimethylol propane, and 2-amino-2-methyl-1,3-propanediol (AMPL) materials. These molecules absorb a great amount of energy through reversible breaking and the formation of nearest neighbor bonds (H-bond) in the molecular crystals. The raise in phase-change enthalpy is mainly attributed to the breakage of square bridge-type molecules present in an H-bonding interaction when the temperature climbs to its transition temperature. Phase-change phenomena of polyalcohols

are usually experiencing a first-order transition and their Gibbs energy changes are approaching zero.^[115,122,123] Thus, these materials have significant potential for passive heating in buildings, that is, domestic water heating systems.

Regarding this, the energy storage performance of pristine polyalcohols like PE, NPG, TAM, and aminoglycol (AMPD) and their multicomponent or binary blends as S–S PCMs for thermal energy was experimentally studied by Gao et al.^[124] with the support of calorimetric analysis. They revealed that pristine AMPD is not practically suitable for emerging TES systems because of its hygroscopic identity and wide-ranging charging and discharging temperatures. However, pristine PE has potential for low-temperature solar TES applications. In addition, PE/AMPD mixtures with a low or high weight ratio of AMP can also be considered promising mixtures for functional applications. In another study, the TES performance of several monobasic, binary, and ternary systems which mainly comprised NPG, PE, and TRIS arranged in varied fractions of masses were expressively studied by Yan et al.^[117] It was reported that the latent heat of reported mixtures of different polyalcohols like NPG/PE and NPG/TAM/PE, had reasonable thermal storage capacity, but relatively lower than pure polyols. For this, various inorganic salts, that is, magnesium chloride or sodium sulfate are being employed to enhance the enthalpy of these materials.^[125] But, the problem is that, such composite PCMs exhibit S–L phase transition instead of S–S phase transition with the practice of these kinds of salts. In this scenario, the use of symmetric dialkylammonium salts is an exciting substitute for both organic and inorganic constituents. Such compounds display S–S phase transition behavior with high latent heat 102–185 J g^{−1} at transition temperature around 20–95 °C.^[126] This performance makes it a promising candidate for a wide array of energy-storage applications with long-term stability. Further, these materials are far more exceptional than conventional PCMs toward solid-state refrigeration technology due to their suitable phase transition temperature and workability within a solid-state operation. In this regard, the adaptability of PCs has been gaining enormous attention in recent years for solid-state coolant sectors to enhance the sustainability and service life of the system. PCs are primarily developed by a certain kind of molecules or ions that offer unique characteristics; 1) long-range order of transition with molecular re-orientational freedom and 2) exceptional mechanical plasticity. In short, PCs are termed as orientationally disordered crystal materials that lie at the borderline between solid and liquid phases. These materials usually involve a crystal to the plastic phase transition during its crystal lattice with the exhibition of large enthalpy and reversible entropy changes. The magnitude of this transition is mainly attributed due to the detergency of the numerous accessible arrangements upsurges by the ordered crystal state to the highly disordered PCs state because of their loosely packed crystal. Thus, the PCs have been extensively used as solid-state TES materials owing to the unique re-orientational motion of molecules or ions during phase transitions.^[127] Particularly, PCs could be provided with additional functionality to design a novel type of multifunctional material that exhibits various changes in response to different channels, that is, magnetism, optics, and electricity in the coexistent system.^[128,129] The excellent report on the state-of-the-art works, not only validates the

multipurpose nature of the PCs but also explained their adoption for diverse applications in the cooling sector.

Adaptability of PCs for Cooling Systems: The adaptability of PCs is growing exponentially for cooling systems owing to the barocalorics^[127,130] and optics^[131] effect in molecular crystals. Order-to-disorder reversible phase transition that can be activated through external effects, that is, temperature, tiny pressure, the magnetic, or electric field has become an attractive area of research because of their importance to enthalpy changes.^[130,132] The working phenomena of caloric materials typically involve phase transition driven by stress applied or removed on the material in terms of elastocaloric effect, electric field (electrocaloric effect), and magnetic field (magnetocaloric effect).^[127,130] Contrarily to the abovementioned caloric effects, the barocaloric effect (BCE) offers the reversible phase transition on the application of external pressure. “Baros” is the Greek word relating to pressure and “caloric” means actions expressing heat. So, the heating and cooling effects in a material influenced by the variation in pressure are known as “BCE.”^[127] The large reversible change in isothermal entropy and adiabatic temperature is necessary for a material to accomplish effective caloric effects. Further, BCE has been implemented in several systems, that is, organic–inorganic hybrid,^[133] shape-memory alloy (SMA),^[134] and PCs as well.^[127,132,135] devising its greatest universal materials among these solid-state caloric effects. Herein, we only focus on BCE effects on PCs-based systems, and other materials are covered in the relevant sections. Recent developments show that PCs have vast potential to achieve the BCE as an alternative solution for the cooling system as compared to conventional vapor-based refrigeration technologies that have harmful volatile hydrofluorocarbons fluids. Thus, PCs with stimulating BCE have gained enormous attention for future eco-friendly refrigeration technologies.^[127,130] For example, they exhibit a great change in isothermal entropy and adiabatic temperature changes from ordered to disordered phase transitions rather than conventional ordered crystals.^[33,127,136] Particularly, BCE effects can be recognized in any sort of TES materials through hydrostatic pressure as it is ubiquitous in nature. The detailed thermo-physical properties of several polyalcohols-based PCMs are highlighted in **Table 2**.

Concerning this direction, several excellent results based on the PCs that include PG, NPG, AMP, and some other closely linked compounds were individually studied by Bing et al. and Lloveras et al. (**Figure 6a**).^[127,130,132] The NPG molecule (chemical formula; C₅H₁₂O₂) usually comprises five carbon atoms that can form the tetrahedron geometry in the crystal lattice. In the designed structure, two carbon atoms are linked to H-atoms in the CH₃ group whereas two others form the hydroxymethyl group. NPG molecules usually crystallize in a monoclinic framework “P2₁/n” with sufficiently arranged molecules having only one lattice configuration at room temperature. The ordered phase changes into a face center cubic lattice (*Fm*₃*m*) at 314 K on heating having 60 configurations. This configuration behavior involves combinations of lattice changes, that is, rotational and conformational degrees of freedom over the ordered structure transition between $\alpha \rightarrow \gamma$ transformations of NPG materials (**Figure 6b**).^[135] Moreover, the arbitrary rotation of the NPG molecules will sustain the FCC proportion around their center of masses and dynamical averaging over

Table 2. The overall summary of polyalcohols-based PCMs for energy storage applications.

PCM	T_m [°C]	ΔH_m [J g ⁻¹]	T_c [°C]	ΔH_c [J g ⁻¹]	Application	Reference
PE	187	313	–	–	Heat recovery process	[137]
PG	81	192	–	45	TES	[121]
NPG	43	131.1	–	–	Flexible electronics	[130]
TAM	134	285	–	–	Heat recovery process	[117]
TRIS	133	245	–	26	TES	[127]
AMPL	84	201	–	–	TES	[35]
Erythritol	118	346	–	–	TES	[138]
Sorbitol	96.1	165.2	–	–	TES	[138]
Mannitol	168	281.3	–	–	TES	[138]
PE/TAM	186	209	159	213	Solar heat storage	[124]
AMPD/PE	181	203	150	183	Solar heat storage	[124]
NPG/TAM	38	27–143	23–30	34–150	TES	[115]
NPG/PE	37	18–68	25–31	19–53	TES	[115]
NPG/TRIS	119	150–227	–	–	TES	[113]
NPG/TAM/PE	118	123	173	27	TES	[124]
PE/PG	82–186	140–270	–	–	TES	[123]
PG/NPG	24–90	66–140	–	–	TES	[123]

time, which is chiefly accountable for the adsorption of energy that is stimulated by thermal agitation. The structural phase transformation can also be attained reversibly by employing a less than 199 MPa pressure around 318 K which is further confirmed by using the results of molecular dynamics simulations (Figure 6c,d).^[127] The stationary and dynamic nature of phases of NPG molecules during phase transformation temperature was analyzed in detail through various latest computational techniques as well as experimental methods.^[127,130] The temperature–pressure phase diagram recording ordered-to-disordered reversible ($\alpha \rightarrow \gamma$) phase transition was done by using high-pressure differential scanning calorimetry (HP-DSC) and indicated a small thermal hysteresis around 14 K (Figure 6e). The notably anharmonic lattice dynamics substantially increase the entropy of the PCs phase. In addition to the entropy of PCs, the (HP-DSC) measurements of NPG displayed a huge change in entropy around (384 J kg⁻¹ K⁻¹) by varying the pressure between normal around 0.1–90 MPa at room temperature.^[127,130] The study signifies that the highest entropy variations were done by a pressure change lower than about 40–45 MPa. PG revealed an even better entropy change of 478 J kg⁻¹ K⁻¹ with analogous parameters, but, the problem is that such PCMs usually involve a larger working temperature around 348 K instead of NPG. The AMP-based system also presented remarkably high entropy changes around 628 J kg⁻¹ K⁻¹ with poor reversibility. However, relatively higher functioning pressure (600 MPa) and operational temperature around 360 K are required.^[132] Furthermore, the values of both enthalpy and entropy change revealed in these PCs are approximately ten times higher than the other influential caloric materials (Figure 6f–h).^[127] Consequently, NPG is being considered as a promising candidate for instant real-life applications due to colossal entropy changes over multiple operational cycles within low dynamic pressure near room tempera-

ture. Further, these studies recommend that PC have only one major reversible phase transition from ordered to disordered with significant entropy changes in a confined range of temperature, which are more favorable for accomplishing the BCE effects. The dynamics of an anharmonic lattice are supported by an intermingling of both conformational and orientational degrees of freedom of molecules, leading toward high BCE in mechanically soft PCs. The mechanical integrity of barocaloric materials over a wide number of compression cycles is very crucial for advanced technical applications, as mechanical compliance is an important feature when considering the practical scope. The abundant accessibility of literature regarding PCs shows the phenomenal potential of these materials in solid-state refrigeration technologies.

In summary, NPG is the most suitable candidate for solid-state refrigeration among others, as it operates at a temperature in the range of 40–45 °C, near ambient temperature with significant entropy changes and decent latent heat in a confined range of temperature. Further, these materials are also being applied in cold storage systems to improve the cooling performance of certain devices through superior heat transfer mechanisms by recovering the waste or exhaust heat. The exhaust of a cooling system usually works in the temperature range between 30 and 50 °C, which means the most suitable candidate is NPG. Thus, these materials can also be used as an auxiliary functional unit for refrigerators and cold storage systems. In this regard, they prevent food items from spoiling at the time of power shut down owing to the heat-storage characteristics of S–S PCMs, which are placed into the fitted column of a refrigerator. So, it is understood that the effective utilization of waste heat from cold storage systems by using these kinds of PCMs allows for minimizing environmental pollution and energy consumption. Further, the installation of S–S PCMs panels may also be placed into a refrigerated truck to keep food

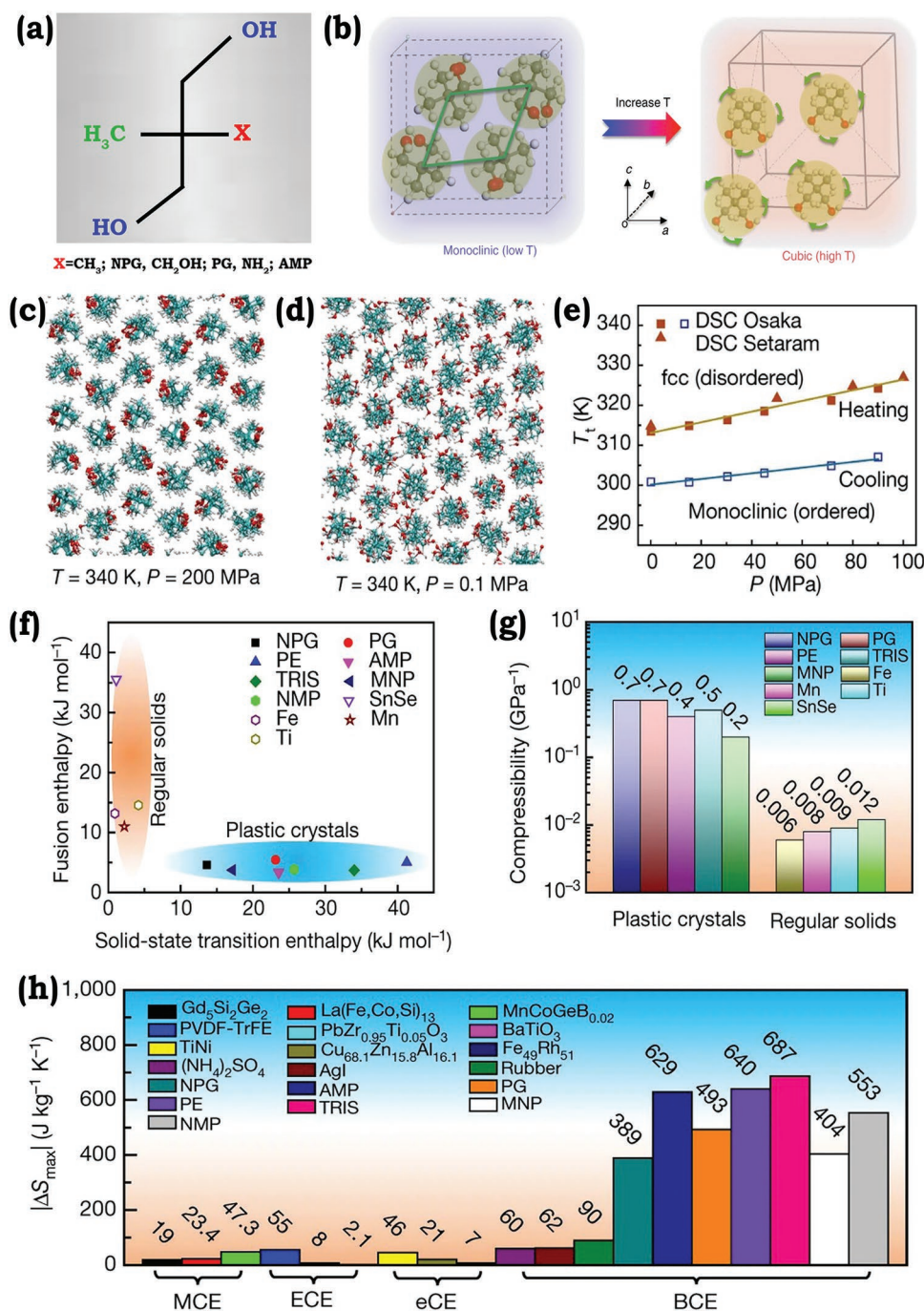


Figure 6. Comparative study of several types of caloric crystals materials examined for next-generation toward solid-state coolants for refrigeration technologies. a) Molecular structures of PCs. (a) includes materials from refs. [127] and ref. [130]. Adapted with permission.^[127] Copyright 2019, The Authors, published by Springer Nature. Adapted under the terms of the CC-BY Creative Commons Attribution 4.0 International license (<https://creativecommons.org/licenses/by/4.0>).^[130] Copyright 2019, The Authors, published by Springer Nature. b) Crystal structure of NPG molecules transforms from monoclinic to cubic phase from low to high temperature. The light-yellow painted circle signifies one NPG molecule, whereas the green mark in the monoclinic phase and green arrows in the cubic phase validates the static and conformational rotation of the NPG molecules respectively. Adapted under the terms of the CC-BY Creative Commons Attribution 4.0 International license (<https://creativecommons.org/licenses/by/4.0>).^[135] Copyright 2020, The Authors, published by Springer Nature. c, d) Molecular dynamics simulations display the structure of the NPG molecules at 340 K under ambient (0.1 MPa) and 200 MPa pressure respectively. e) Temperature–pressure phase diagram analyzed by high-pressure DSC. f) The graph between fusion enthalpy versus solid-state transition enthalpy for regular solids are represented by orange shading while blue regions indicate PCs; PG, AMP, PE, NPG, TRIS, NMP (2-nitro-2-methyl-1,3-propanediol), and MNP (2-methyl-2-nitro-1-propanol). g) Compressibility of representative regular solids and PCs. h) Absolute values of significant entropy change “ ΔS_{max} ” for achieving effective caloric materials. PVDF-TrFE refers to poly(vinylidene fluoride-co-trifluoroethylene). c–h) Adapted with permission.^[127] Copyright 2019, The Authors, published by Springer Nature.

items frozen for a longer time without requiring external power sources in a passive cooling system.

3.1.3. Organic Salts

Organic salts are relatively a new kind of material in the family of S–S PCMs for renewable energy storage applications due to their intrinsic advantages including chemical thermal stability, and economic viability with non-flammable and environment-friendly nature. In addition, the crystal structure of organic salt is an extensive spectrum regime of cations and anions that offer the chance to adapt to target applications by increasing the number of hydrogen donor and acceptor sites via tuning the melting temperature. Therefore, it's been considered a favorable candidate for the potential of low-cost solar TES and passive heating, especially for domestic purposes.^[139,140] Despite these advantages, these salts usually exhibit relatively lower phase-change enthalpy in the case of S–S phase transitions rather than S–L phase transitions limiting their practical applications

to some extent. Regarding this, the well-known MacFarlane group developed a series of organic salts or protic ionic salts for renewable TES applications. Examples of these organic salts; [Pzy][CF₃SO₃], [Gdm][CF₃SO₃], and [Gdm][C₂H₅SO₃], that exhibits S–S phase transition among the family of “Pzy” and “Gdm” salts including [Pzy][CH₃SO₃], [Gdm][C₆H₅SO₃], and [Pzy][C₆H₅SO₃].^[140–144] It was revealed that each of the salts displays a distinctive crystal structure in the solid phase, where electrostatic interactions, that is, Coulombic and H-bonding interaction between cation and anion are being the fundamental driving force for packing of crystal structure. Thus, the TES performance of these materials is mainly relying on the H-bonding within the molecular origins of the crystal structure. From the [Pzy] family, [Pzy][CF₃SO₃] is the only salt that exhibits solid-state transition for renewable TES as mentioned above. However, they exhibit a relatively lower ΔH_f value due to the electron-withdrawing group “CF₃” present in the organic salt (Figure 7a).^[140] Due to the electron-withdrawing effect, electrostatic interaction between cations and anions is present at longer distances, which weakens the H-bond in the crystal

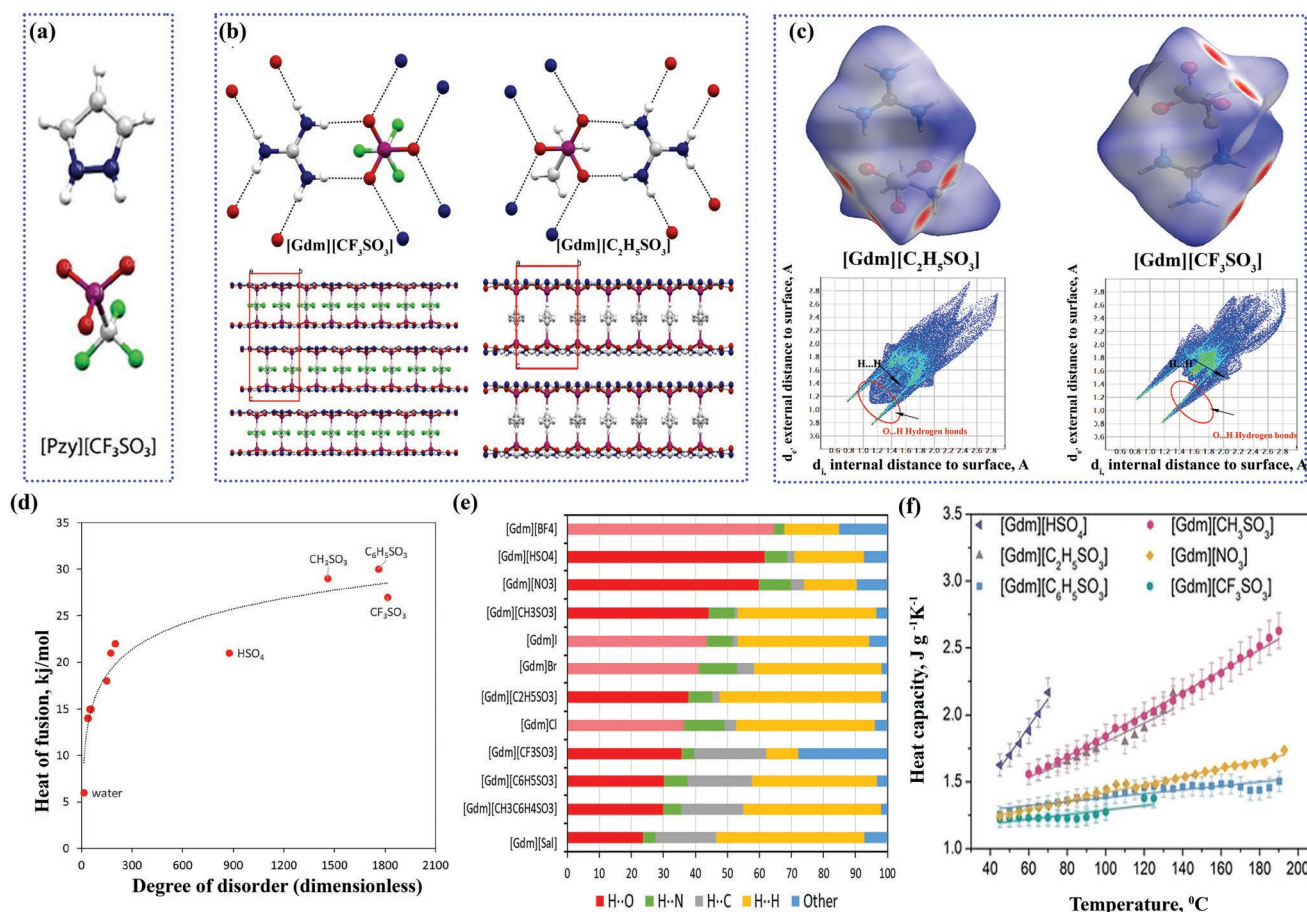


Figure 7. Organic salts for renewable TES applications. a) Crystal structure of [Pzy][CF₃SO₃]. Adapted with permission.^[140] Copyright 2020, Wiley-VCH. b) Crystal structure and lattice structure of [Gdm][CF₃SO₃] and [Gdm][C₂H₅SO₃] with one ion pair and hydrogen bond structure, respectively. c) Hirshfeld surface analysis was done to quantify the intermolecular interaction for guanidinium compounds. The red region corresponds to a hydrogen-bonded acceptor with a neighboring one, while the blue color corresponds to the 2D fingerprint plot with all interactions in terms of N—H...O for guanidinium compounds. d) Relationship between ΔH_f and degree of disorder. e) Percentage contribution of intermolecular interactions in [Gdm]-based organic salt. b–e) Adapted with permission.^[142] Copyright 2019, American Chemical Society. f) Specific heat capacity (C_p) [Gdm]-based organic compounds with the influence of temperature in the range from 45 up to 20 °C below the melting point. Adapted with permission.^[143] Copyright 2021, Wiley-VCH.

structure among the other salts that led to lower phase-change enthalpy.^[140]

Later on, Gdm-based salt was also studied by a similar group for renewable TES applications.^[142,143] They observed that each crystal structure of Gdm salts, that is, $[\text{Gdm}][\text{CF}_3\text{SO}_3]$ and $[\text{Gdm}][\text{C}_2\text{H}_5\text{SO}_3]$ has a unique motif in the solid phase. These salts usually have six hydrogen bonds linking with three $[\text{Gdm}]^+$ as cation with three anions and vice versa for creating a 2D hydrogen-bonded layer sheet (Figure 7b).^[142] Even though, these sheets are not completely planar due to the different kinds of interaction present in their crystal structure. Hershfield surface studies for $[\text{Gdm}]$ -based salts exhibited that, the surface color is shaded from blue $\rightarrow$ red presenting distances between atoms inside the surface is near to one from outside. The darker blue color is visible on the surfaces indicating the presence of long-distance within the atoms. In turn, specifically close interactions are red-colored and they found that these red spots have different sizes validated by two fingerprint plots, that is, $\text{O} \cdots \text{H}$ bond interactions (Figure 7c).^[142] These features exhibited that the crystal structure of guanidinium-based salts was mainly based on the nature and extent of hydrogen bonding interactions. The measured curve of the heat of fusion against the degree of disorder suggests a vibrant relationship was observed in the salts. The disordered magnitude of these compounds looks the same in the liquid phase when all of the H-bonding linkage ($\text{N}-\text{H} \cdots \text{X}$) is interrupted on melting (Figure 7d,e).^[142] Though, the materials with angles nearer to linearity resulting relatively improved enthalpy (ΔH_f) with latent heat around 28 kJ mol^{-1} . In addition, the Gdm cation is a more promising salt for potential enthalpy of melting owing to the multiple hydrogen bonding sites. The quantity of specific heat capacity (C_p) concerning temperature difference was calibrated by DSC data. It was noted that the C_p varies with the anion and increases in the following order $[\text{CF}_3\text{SO}_3]^{-1} < [\text{NO}_3]^{-1} < [\text{C}_6\text{H}_5\text{SO}_3]^{-1} < [\text{CH}_3\text{SO}_3]^{-1} < [\text{C}_2\text{H}_5\text{SO}_3]^{-1} < [\text{HSO}_4]^{-1}$ (Figure 7f),^[143] which is mainly based on the melting temperature of salts. Thus, ΔH_f mainly depends on hydrogen bonding present in the crystal structure, being a function of not only their number but also the strength of the hydrogen bonds (length and deviation from linearity).

3.2. Inorganic-Based S–S PCMs

Inorganic-based S–S PCMs are a kind of PCMs that are capable to store or release thermal energy through various switching mechanisms including order–disorder transformation, crystallographic structure transformation, and magnetic transformation from crystal structure to an amorphous structure. Broadly speaking, the extent of phase-change enthalpy of inorganic-based S–S PCMs is usually lower than organic and polymeric S–S PCMs with high phase transition temperatures. Regarding this, a variety of PCMs with S–S phase transformations was studied for achieving a reasonable amount of latent heat or discharge heat at a specific temperature.^[51,52,145–148] It was found that magnetic transformations and crystallographic structure are the most widely studied transition among other transitions owing to their intrinsic switching of crystal structure. Nevertheless, such kinds of materials are quite scarcely reported in

the literature survey, which can undergo multifold and reversible S–S transformation at narrow temperatures. Besides this, the most adaptable application of inorganic S–S PCMs is in industrial power plants to recuperate heat from the irregularly discharged heat or exhaust gases. The basic phenomena of inorganic S–S PCMs in an industrial power plant involved stimuli controlled phase change at high temperatures to store latent heat energy significantly through thermal charging of waste and later on release it during solid-state operation. Generally, inorganic S–S PCMs can be classified into ceramics, metallics, and metal alloys based on their materials composition.

3.2.1. Ceramics

Ceramics are those materials that are neither organic nor metallic. It may be crystalline, glassy, or exhibits both crystalline and glassy states of matter. In addition, they are typically stiff and chemically non-reactive materials that can be designed for use under high temperature and high heat flux without too much consideration of the serviceability. Recently, ceramics-based heat-storage materials have been gaining substantial attention in terms of energy-saving applications, as they can conserve latent heat energy for a longer time and release accumulated heat on demand by the application of minute pressure. The development of high-performance heat-storage materials is significant for the adequate use of exhaust heat/waste heat coming from blast incinerators of power plants and factories. In this regard, inorganic S–S PCMs are considered to be convenient as latent heat storage materials owing to their significant characteristics. These are further divided into two categories: 1) inorganic salt for high-temperature thermal storage and 2) long-term heat-storage ceramic-based materials for recycling waste heat.^[51,145,149] In the first case, inorganic-salt-based S–S PCMs have been developed as a promising aspirant for high-temperature TES applications. For this, lithium sulfate is significantly used because it undergoes S–S phase transformations from monoclinic to FCC at a very high temperature of around 576°C , which is most suitable for concentrated solar thermal (CSP) technologies.^[51,148] Nevertheless, this single material is not adequate for supercritical CO_2 power cycles, as it has a high cost and requires high energy density over a narrow temperature range. To overcome such issues, cascaded-based PCMs systems are built for CSP technologies. In this respect, the combination and mixture of lithium and sodium sulfate with (40% Li_2SO_4 :60% NaLiSO_4) ratio to build an S–S cascaded PCMs with higher TES density was explored by Bayon et al.^[51] The prime use of these materials is to increase the temperature variations of operations for the sake of the CO_2 power cycle as well as diminish the overall cost of the storage system. For this purpose, they integrated S–S PCMs cascaded system with CSP technologies and found superior benefits over S–L PCMs in terms of no corrosiveness, no encapsulation, low volatilization, etc. The cost-effective assessment result of a single PCM and cascaded system displayed that such a combination of materials has great potential for supercritical CO_2 power cycles with a projected cost of $\$50.2 \text{ kWh}^{-1}$.

On the other hand, frequently existing phase-change heat-storage materials cannot usually store energy for a longer

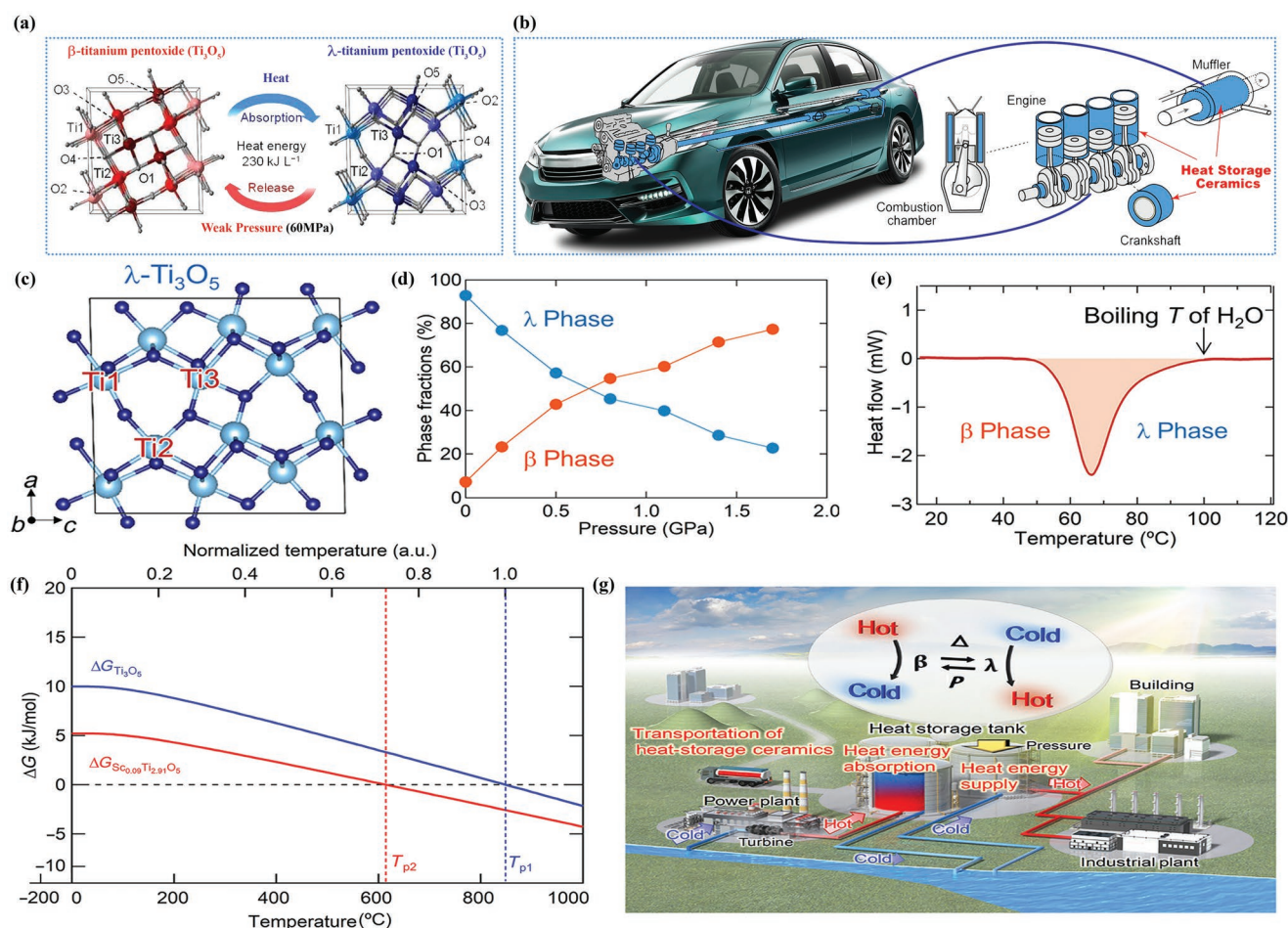


Figure 8. a) Crystallographic structure of strip-based Ti₃O₅ heat-storage materials with their pressure and heat-induced phase switching between λ -Ti₃O₅ and β -Ti₃O₅. Adapted under the terms of the CC-BY Creative Commons Attribution 4.0 International license (<https://creativecommons.org/licenses/by/4.0/>).^[149] Copyright 2015, The Authors, published by Springer Nature. b) Schematic representation of proposed applications of block type Ti₃O₅ heat-storage materials have been applied in different components of automobiles: engine, muffler, and crankshaft. Adapted under the terms of the CC-BY Creative Commons Attribution 4.0 International license (<https://creativecommons.org/licenses/by/4.0/>).^[150] Copyright 2019, The Authors, published by Springer Nature. c) Crystal structure of the Sc-substituted materials λ -Ti₃O₅. d) The phase fractions of Sc_{0.09}Ti_{2.91}O₅ followed pressure dependence measured by the SXRD patterns indicating crossover pressure transition in designed materials occurs at 670 MPa. e) DSC chart of Sc_{0.09}Ti_{2.91}O₅ displays an endothermic reaction experienced at 67 °C. f) Systematic mechanism for the decrease in transition temperature represents a difference in free energy ΔG by means of λ and β phase transitions. g) Schematic illustration of recycling of heat energy from power plant system using Sc-substituted λ -Ti₃O₅ heat-storage ceramics materials for portable applications. c–g) Adapted under the terms of the CC-BY Creative Commons Attribution 4.0 International license (<https://creativecommons.org/licenses/by/4.0/>).^[152] Copyright 2020, The Authors, published by AAAS.

period over the relatively high phase transition temperature due to the heat losses phenomena happening within transitions. From this angle, the current research progress is shifted toward relatively less studied phase transition materials, where the thermal energy can be conserved as latent heat for a prolonged time, for example, seasonal heat storage until the application of pressure is applied. In this way, stimuli-controlled PCMs have been esteemed as encouraging aspirants for mobilized or portable TES applications. In this kind of PCMs, latent heat is stored through thermal charging of waste heat sources and later released upon the demand placed by employing external stimuli regardless of temperature confines.^[150–152] The stimuli-controlled energy conservation in inorganic Ti₃O₅-based PCMs was studied by Tokoro and co-workers. The strip-based

Ti₃O₅ stored heat energy by 228 kJ L⁻¹ over an S–S phase transformation around 258 °C and released heat energy by the application of a tiny amount of pressure at room temperature. The reverse-phase-change transition was activated by heat, electric current, and light (Figure 8a).^[149] In short, this kind of phase transition material reversibly stores/releases heat energy with the help of pressure, light irradiation, and thermal action, which allow latent heat release control and offers several opportunities in the portable TES system. Following similar directions, some additional applications of the existing material should also be viable. For example, heat-storage usage to collect waste heat from factories, vehicles, and electronic devices.^[150] Low-pressure-responsive heat-storage ceramic materials are gaining tremendous attention in this regard. Particularly,

block-type (Ti_3O_5) materials were extensively employed to recover the excessive heat efficiently in automobiles engines and exhaust heat from industrial factories on account of their stimuli-responsive behavior. They also accumulate a great amount of phase-change enthalpy around 238 kJ L^{-1} which is relatively higher than strip-based materials. In addition, such materials can warm the chilled internal system, when an automobile engine resumes. Thus, such materials would be suitable for automobile car segments neighboring engines as well as mufflers, as the activating pressure are less than 10 MPa (Figure 8b).^[150]

Later on, current research has paid much attention to metal substituted-based materials particularly lambda-trititanium-pentoxide ($\lambda\text{-M}_x\text{Ti}_{3-x}\text{O}_5$), where M may be transition metals: Sc, Nb, Ta, Zr, Hf, and W. In this direction, Nakamura et al.^[152] synthesized long-term heat-storage ceramics materials based on scandium substituted lambda-trititanium pentoxide ($\lambda\text{-Sc}_x\text{Ti}_{3-x}\text{O}_5$) via arc melting technique. The developed materials collected heat energy that is produced from thermal and nuclear power plants by hot water on account of the S–S phase transition at $38\text{--}67^\circ\text{C}$ and then liberates heat energy on demand by applying a small amount of pressure (60 MPa). Herein, it was revealed that the temperature of the heat energy should be lower than the boiling point of water. The blue elements in the crystal structure of ($\lambda\text{-Sc}_x\text{Ti}_{3-x}\text{O}_5$) exposed lower energy than the pure materials, while the orange color presented higher energy (Figure 8c).^[152] The as-prepared material was compressed via hydraulic press using pressure from 0.2 to 1.7 GPa. The crossover pressure fraction displayed that the λ phase fraction of the materials decreased by increasing the pressure and reversed trend was observed in the β phase fraction (Figure 8d).^[152] After the pressure induced, the heat-storage property was observed at a suitable phase transition of 67°C by the conversion between β and λ phase with enthalpy ΔH of 75 kJ L^{-1} (Figure 8e). The decrease in heat-storage temperature is mainly recognized by lessening in the formation of energy differences between two phases that ultimately diminishes the crossing temperature of two Gibbs energy curves accompanied by first-principle calculations (Figure 8f).^[152] It is worth mentioning that the developed materials can maintain their good stability for a prolonged period even after 367 days nearly 1 year because of the presence of an energy barrier between β and λ phases. Additionally, heat absorption by this material below (100°C) can recuperate thermal energy from the cooling water in power plant turbines and eventually help to diminish the increase in seawater temperature. The stored thermal energy by this material can be delivered to industrial plants and buildings, which are near power plants irrespective of the use of electricity (Figure 8g).^[152] This means that the stored heat can be delivered to a specific location by truck, that is, houses as the latent heat energy hold for a prolonged time, until pressure applied.

Conclusively, a sustainable heat recovery system governed by controlled heat release PCMs has been developed being a promising candidate for energy-saving and reuse of this waste heat efficiently for end users. The core parameters including a suitable working temperature range of $35\text{--}65^\circ\text{C}$ with reasonable latent heat storage capacity and stored heat energy for a long time are highly desirable for waste heat recovery applica-

tions. Long terms ceramics-based materials are suitable for this purpose. These materials could also be suitable to recover waste heat from vehicles engines and electronic devices. This discussion shed light on further investigation of new heat-storage materials for eco-friendly energy reuse technology by using waste heat of thermal and nuclear power plants.

3.2.2. *Metallics and Metal Alloys*

Metallics and metal alloys are usually defined as a class of metallic S–S PCMs that are typically employed as solid-state TES applications following reversible martensitic phase transition. Regarding this, NiTi-based alloys gained significant consideration owing to their reversible martensitic transformations. Further, these materials have several sealant advantages, including high strength, ductility, exceptional corrosion resistance, and decent formability through conventional thermomechanical processing.^[146,153–155] Herein, J. Sharar et al. explored the use of NiTi-based binary, ternary, and quaternary SMA smart materials as an ultra-high performance of metallic solid-state PCMs for cyclic TES.^[146] It has been reported that the S–S PCMs crystalline phase conversion was observed during the metallic alloys phase transition between the low-temperature monoclinic B19[−] Martensite (M) structure phase to the high-temperature cubic B₂ Austenite (A) lattice configuration. The shape memory alloy NiTi stored latent heat energy up to (36 J g^{-1}) through an S–S phase transformation temperature ranging from -50 to 500°C . Moreover, the unique properties like latent heat, phase changing temperature, and structural properties of SMA materials can be triggered by varying the equilibrium between Ni and Ti via introducing trace elements. This magnitude helps to improve the design and operation flexibility of electronic devices. Consequently, SMA is being considered as a potential candidate in electronics applications. Regarding this, some chalcogenide-based alloy compounds are also being employed in nonvolatile memory devices neuro-inspired computing for their dynamic role in data storage and processing. In this regard, Ge–Sb–Te-based alloys compounds such as $\text{Ge}_2\text{Sb}_2\text{Te}_5$, InGeTe , GeSb_2Te_4 , and $\text{Ge}_{13}\text{Sb}_7\text{O}_{16}$ are notably applied for data storage applications in memory devices due to disorder-metal-insulator transitions between a crystalline and amorphous state of PCMs.^[156–158]

Apart from heat storage, inorganic S–S PCM also exhibits a thermochromic effect in respect of optical modulation. Thermochromism is another form of stimuli-responsive behavior in molecular single crystals that exhibit a difference in color or light emission through a thermally induced phase transition. Because of the reversible thermal-induced phase transition, several crystalline materials can also be favorable as temperature sensors and latent heat storage units.^[159,160] In this context, VO_2 exhibits a strongly interconnected electron system that usually shows a crystal phase transformation between the monoclinic phase to tetragonal geometry around the phase transition at 68°C . The simultaneous change in the origin of VO_2 is typically observed through the degree of freedom following metal-to-insulator transition or MIT spin transition. VO_2 was widely studied as inorganic PCMs for thermochromic smart window fabrication or coating to reduce the overall energy consumption of buildings.

Therefore, a thermo-stimuli response-based window is being considered as an efficient attitude toward smart windows because of its automatic switching in terms of energy-saving and energy storage. It is a much more realistic and practicable approach to achieving human demands, as it can be passively triggered over the weather-changing state.^[161] This kind of PCMs is normally packed between two glass plates to thermally insulate window structures against daytime temperature variations in terms of window application.^[162] The PCMs integrated window store's heat during the day by melting the materials and converting windows into optically transparent behavior; later on, this stored heat is released at night to carry out thermal management solutions. In addition to a thermochromic application, inorganic materials also presented an intrinsic change in their transparency behavior over phase transition switching. Due to this property, it may be more convenient to design low-cost S–S PCMs systems for advanced thermochromic applications at a broader scale. The energy efficiency of a building can be improved by manipulating window structures through thermochromic PCMs for optical modulation. During winter, the S–S PCMs are crystalline and present opacity at ambient temperatures, and transmit solar radiation through the external surface of the window resulting in lower heating demands. Conversely, during summer, the S–S PCMs perform amorphous characteristics and become transparent at higher temperatures. Due to this, solar radiation is reflected by a thermochromic window leading to lower internal temperatures. Contrary to this, heat-storage-based PCMs do not display valuable properties in their thermochromism structures. Thus, they are not an appropriate choice for window applications because of the deficiency in light spread control. Nevertheless, these materials are acceptable for passive temperature management of thin window structures by means of thermal insulation.^[163]

3.3. Hybrid S–S PCMs

Hybrid S–S PCMs, a relatively new class of PCMs that are mainly composed of organic/inorganic, coordination of metal cations and anions, have earned significant attention for various thermal-energy applications due to their fascinating properties, that is, high thermal stability and non-flammability. Therefore, these materials can be extensively applied for both solar thermal and electric energy storage because of the existence of diverse functional properties like dielectric, optical, and caloric. Although, these materials generally present relatively lower latent heat than most of the S–S PCMs. Besides this, some of the hybrid materials including spin-crossover (SCO) complex and perovskite-oxide-based compounds have gained significant potential for solid-state barocaloric refrigerants systems due to a suitable phase transition temperature in the range of 44–63 °C with reasonable latent heat. The phase transition of these compounds can be tuned by varying the mole fraction of its chemical substitution. Further, these compounds can also be suitable for various solid-state devices with a range of operating temperatures. Herein, we will briefly cover organometallic and perovskites-based complex compounds following reversible S–S phase transitions toward potential thermal-energy applications.

3.3.1. Organometallic S–S PCMs

Organometallic S–S PCM at present mainly refers to a layered perovskite-organometallics, mainly constituted through alternative layers between organic/inorganic constituents resulting in a sandwich-like crystalline structure. These materials are usually designated as numerical formula $(nC_nH_{2n} + 1NH_3)_2MX_4$, where M presents a metal atom, X shows a halogen group, and n is the number that typically ranges from 9 to 19 as explained earlier, in which, the organic segments comprised of alkyl chains of n -alkyl groups are chemically bonded with the inorganic layer via ionic bonds. The phase transition of these materials mainly includes crystallizing motif chains, whereas, the inorganic components have remained structurally unaffected because the supporting framework is thermally inactive. The crystallizing group is ordinarily melted and undergoes phase transformation phenomena between an ordered state or crystalline structure to a partly crystalline or disordered one over absorption of heat energy. This means that the phase-change enthalpy and melting temperature of these materials are mainly based on the chain length of crystallizing motif and partly based on specific metal and halogen identities within an inorganic group. Nonetheless, inorganic constituents mainly involve the improvement in weight aptitude of organometallic S–S PCMs rather than latent heat capacity. Therefore, the latent heat storage capacity of these compounds per unit mass is relatively lower as compared to the melting enthalpy of normal S–L PCMs. Generally, the S–S phase transition of the hybrid materials exhibits reversible behavior even after multiple cycles with negligible change in volume roughly 5–10% at solid-state transition.^[28,56] In this context, several organometallic S–S PCMs have been prepared and categorized employing $X = Cl$, with a melting enthalpy range from 63.5 to 152.7 J g^{−1} following the S–S phase transformation temperature between 20.1 and 95.2 °C.^[164–167] Among them, thermal characteristics of Co- and Zinc-based binary combinations were perceived up to mark. The binary combinations of these materials usually presented similar performance to priorly studied polyalcohols blends for carbon atom (C9–C15) at transition temperatures of 75 and 100 °C with latent heat enthalpy between 73 and 110 J g^{−1}.^[166]

Recently, hybrid halometallates are growing as a favorable candidate means of combining TES with multifunctional designs for new high-tech applications as they have shown better thermal stability with non-flammability behavior over traditional S–L PCMs. In addition, the family of hybrid materials also offers different options for new TES systems based on hybrid porous materials. For instance, evaporative TES systems are designed through metal–organic frameworks.^[168,169] Herein, Salgado-Beceiro et al. explored a novel hybrid halometallate $[(CH_3)_3S][FeCl_4]$ as an innovative material for multidimensional-energy storage system about TES and electric energy storage with PCs transition behavior.^[170] This material experiences a first-order solid-state plastic transition at ambient temperature with decent enthalpy ≈ 40 kJ kg^{−1} with a transition temperature range between 42 and 29 °C, which is generally suitable for solar TES. It has been revealed that such kinds of material also exhibit multidimensional properties with temperature-induced reversible transformations in its dielectric, magnetization, and electron spin resonance manners associated

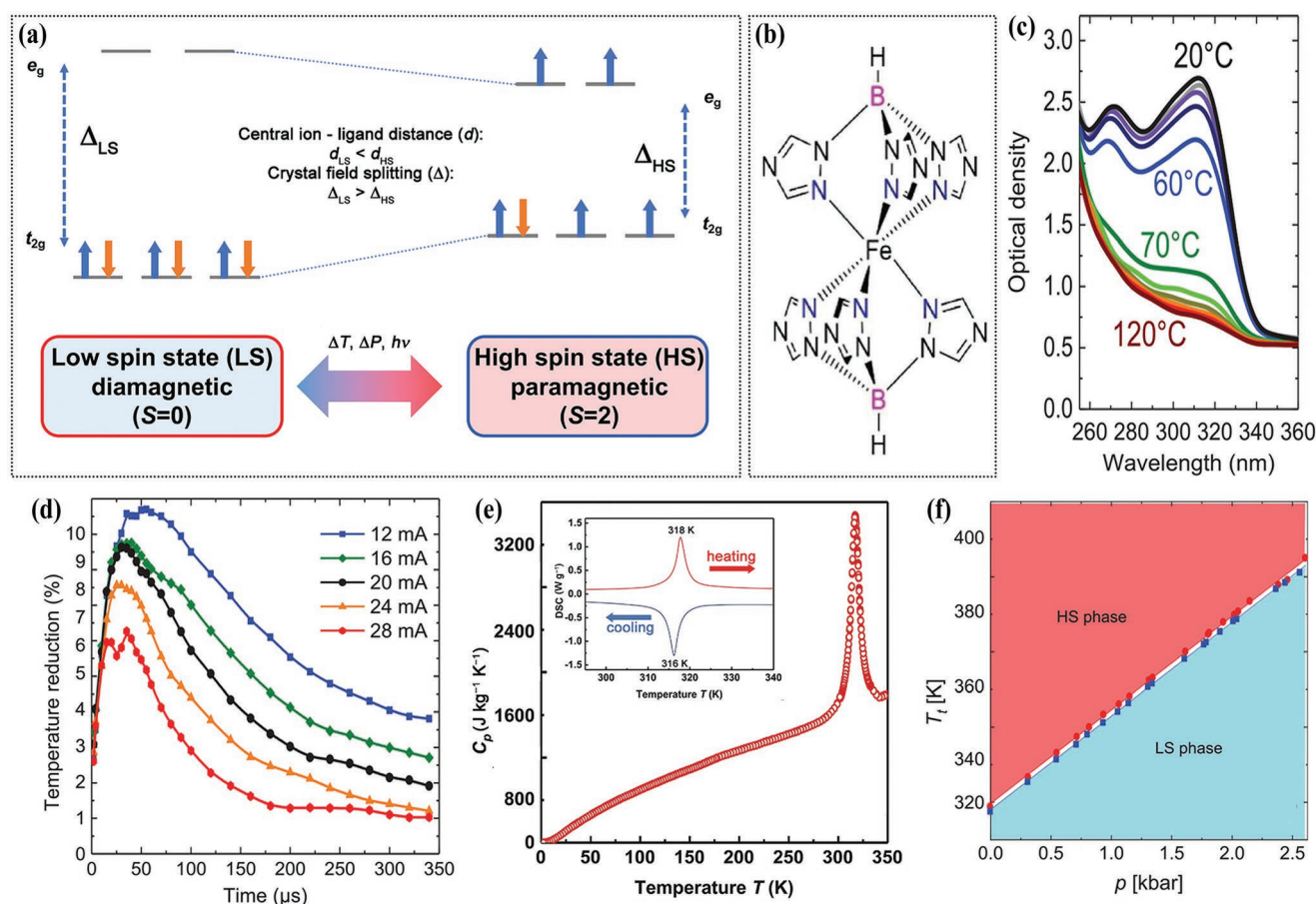


Figure 9. a) Schematic diagram of SCO metal ion of transition elements with d_6 electronic configuration. Adapted with permission.^[173] Copyright 2021, Wiley-VCH. b) The molecular structure of thermally induced spin-crossover (SCO) transition metal complex compound is entropy-driven for thermal-management applications. c) Optical absorbance spectra of the film were acquired at several highlighted temperatures in the heating mode between 20 °C (LS state) and 120 °C (HS state). d) Time evolution of the percentage of temperature reduction κ (%) in the wire, following the application of a sharp current step of different intensities. b–d) Adapted with permission.^[172] Copyright 2020, Wiley-VCH. e) DSC analysis in terms of specific heat versus temperature. The upper and lower curve in the inset corresponds to endothermal LS $\rightarrow$ HS transition on heating and exothermal HS $\rightarrow$ LS transition on cooling. f) Temperature-dependent isobaric heat flow in terms of transition temperature as a function of hydrostatic pressure. The blue signs exhibit HS $\rightarrow$ LS transition whereas red signs present opposite transition from LS $\rightarrow$ HS. e, f) Adapted with permission.^[173] Copyright 2021, Wiley-VCH.

with phase transition. Therefore, this compound is extremely stimulating for achieving both solar thermal and electric energy storage owing to its coexistence and multifunctional properties. In short, solar radiation can be transformed and conserved as thermal energy by the designed material which brings a significant rise in the magnitude of electric energy that can be stored through capacitors built by this material because of their dielectric permittivity property.

Lately, molecular SCO complex compounds have been recognized as an outstanding contenders in solid-state barocaloric refrigerants.^[169,171–173] The examples of SCO complexes include octahedrally coordinated 3d transition metal ion materials that can undergo a reversible transition between two electronic arrangements, low-spin (LS) and high-spin (HS) states on account of their minimum and a maximum number of unpaired d-electron, respectively. The interconversion phenomena between these two states can be activated with the help of external stimuli (change in pressure or temperature and light irradiation, etc.). The antibonding identity of e_g orbitals signifies definite flagging and broadening of the metal–ligand

interaction in the high spin phase. The higher-spin diversity ($2S + 1$) ultimately lowers the latent energy of the high spin state improving magnetic as well as vibrational entropy that led to the low spin $\rightarrow$ high spin switching over relatively increasing temperature. However, a slight upsurge in pressure helps to reduce the volume at a low spin state over a sudden transition that led to a substantial decrease in entropy, which can be harvested as a caloric effect (Figure 9a).^[173] Further, the potential value of SCO complexes concerning thermal characteristics has been moderately explored for barocaloric refrigeration technology to resolve thermal management issues in a functional device. Regarding this, the thermal damping effect of the SCO complex molecules $[Fe(HB(tz)_3)_2]$ ($tz = 1,2,4$ -triazol-1-yl) concerning its different molecular films was studied in a functional device (Figure 9b).^[172] The obtaining nano-crystalline films were revealed to be consistent, steady, and presented a sudden spin transition at a temperature of SCO around 64 °C. In addition, the spectra of optical absorbance for designed film established at distinct specific temperatures were also studied (Figure 9c).^[172] Further, the maximum proportion in the reduction of the

temperature curve was experienced over the function of time due to its heat absorption ability. It was highlighted that the temperature-damping effect usually drives toward a maximum stage, before it start declining with respect to time at a given current ranging from 12 to 28 mA. The analysis curve indicated that the percentage of decrease in the temperature curve was mainly based on excitation current (Figure 9d).^[172] The optimal efficiency in terms of reducing the temperature is not instant, but it normally happens after varied tons of micro-seconds. The observation is mainly attributed to thermal spin-transition in the films of SCO compound that is not entirely abrupt. Conclusively, it is worth mentioning that the rise in temperature in microcircuits can be efficiently restrained through the excessive heat capacity of materials followed by S–S spin-transition. Further, the heat-storage property of the SCO-based thermal damper was found to be completely conserved even after 1000 heating swings in the air. Similarly, recent studies on a colossal and reversible BCE effect in trinuclear SCO compound, that is, $\text{Fe}_3(\text{bntrz})_6(\text{tcnset})_6$ named as FBT have been reported where $\text{bntrz} = 4\text{-(benzyl)-1,2,4-triazole}$ and $\text{tcnset} = 1,1,3,3\text{-tetra-cyano-2-thioethylepropenide}$. The spin state transition is associated with distinct specific heat capacity curves verified through DSC data of the FBT complex. The observed phase-change enthalpy and isothermal entropy changes are $\Delta H = 30 \text{ J g}^{-1}$ and $\Delta S = 118 \text{ J Kg}^{-1} \text{ K}^{-1}$ correspondingly with an adiabatic temperature change of 35 K. The authors' results presented that a large peak association was observed for the FBT complex in both the heating and cooling cycles over transition temperature between 317 and 316 K, respectively (Figure 9e).^[173] Further, the increase in transition temperature was linear with a slope $dT/dp = 25 \pm 0.2 \text{ Kbar}^{-1}$ exhibiting similarity with a temperature of the DSC curve for both low spin $\rightarrow$ high spin transition and vice versa (Figure 9f).^[173] The reason is that the hysteresis of the spin switching is not influenced by hydrostatic pressure. Further, these values are experienced to be reversible over a low pressure or broad temperature range highlighting the possible practical importance of this material in solid-state refrigeration.

3.3.2. Perovskites-Based S–S PCMs

Organic–inorganic hybrid perovskites compounds are typically cost-effective and easy to obtain with an eco-friendly attitude. These materials are usually represented with the molecular formula; A_2MX_4 , where A refers to pliable organic cation, M is metal ion and X may be Cl, Br, I, COO^- , and CN^- . The phase transition study of these compounds is usually escorted by symmetry breaking under the influence of temperature. Hence, hybrid-perovskites PCMs can form crystal structures between 0D and 3D concerning symmetry. Broadly speaking, perovskites or layered perovskites are usually blended with organic/inorganic compounds constituents, to yield a new material. These materials generally exhibited reasonable phase-change enthalpy through a completely reversible phase transitions pathway.^[174,175] They have resembled behavior similar to organometallic or hybrid halometallates, whose functionality is also linked with the structural transition. Due to interesting features like antiferroelectricity, switchable dielectric, and non-linear optical properties, perovskite compounds are becoming an excellent candidate for environmentally friendly solid-state

cooling device applications.^[176–179] Therefore, recently perovskites-based S–S PCMs are attracting significant attention in the field of TES, which can be favorable for solid-state devices with several working temperatures.^[54,165,178,180]

In this connection, Uchimura et al. studied the quadruple perovskite oxides $\text{RCu}_3\text{Fe}_4\text{O}_{12}$ family (R; rare earth metals and it may be La, Pr, Nd, Sm, and Gd) that display variable transition temperatures between $(-19 \text{ and } 95^\circ\text{C})$ for dynamic TES.^[181] These oxides are normally crystallized in the two divergent transition metals referring to copper Cu and Fe, depicting discrete crystallographic sites of pseudo square A- and octahedral B-sites correspondingly from low temperature to high temperature through intersite charge-transfer phase transition (Figure 10a).^[181] In addition to their crystallographic structure, these oxide experiences two divergent sorts of electronic phase transitions including intersite charge transfer (ICT), and charge disproportionation transitions (CD) that rely on the size of the R ion. The definite latent heat of complex perovskites compounds ($\text{RCu}_3\text{Fe}_4\text{O}_{12}$) was measured over a function of temperature. Their finding indicated that transition enthalpy of established perovskites-based PCMs regularly drops from 26 J g^{-1} with substitution of $\text{R} \rightarrow \text{La}$ to 16 J g^{-1} , R; Gd with the lowered transition temperature suggests a decrease in TES performances due to its elemental replacement, that is, $\text{V}_{1-x}\text{W}_x\text{O}_2$ (Figure 10b).^[181] In short, the replacements of R ion in $\text{RCu}_3\text{Fe}_4\text{O}_{12}$ attains the control of working temperature for TES material, as it is decisive for colossal phase-change enthalpy to entropy changes. Nevertheless, a significant reduction was observed in transition entropies over doping of tungsten (W) and chromium (Cr) in $\text{V}_{1-x}(\text{Cr}, \text{W})_x\text{O}_2$. Overall, the author's results endorsed that the replacements of the atoms are implicitly connected to the electronic state of the crystal structure, which helps to keep the TES property and control the transition temperature.

Similarly, a-site-ordered quadruple perovskite-structure oxide $\text{NdCu}_3\text{Fe}_4\text{O}_{12}$ presenting favorable latent heat at ambient temperature owing to the ICT transitions, were explained by Kosugi et al.^[182] The as-prepared perovskite compound usually crystallizes through a cubic ordered quadruple structure following Nd and copper (Cu) are ordered in 1:3 ratio at the A site, while the iron is placed on the center of the BO_6 octahedra. It was also depicted that the inherent changes are constant with an electronic structure changes between the high-temperature state ($\text{Nd}^{3+}\text{Cu}^{2+}_3\text{Fe}^{3.75+}_4\text{O}^{2-}_{12}$) to low temperature one ($\text{Nd}^{3+}\text{Cu}^{3+}_3\text{Fe}^{3+}_4\text{O}^{2-}_{12}$) (Figure 10c).^[182] The latent enthalpy and entropy change obtained through the study of first-order transitions was found to be 26.4 kJ kg^{-1} (158 J cc^{-1}) and $85.3 \text{ J K}^{-1} \text{ kg}^{-1}$, respectively. The ICT transition temperature decreases linearly by 301.5 K at 0.1 kbar to 280.5 K at 5.2 kbar corresponding to the pressure coefficient of the intersite transition. The author's results highlighted that experienced entropy changes in the developed structure through BCE is analogous to previously reported inorganic solid materials (Figure 10d,e).^[182] This fact is mainly attributed to the presence of unusual magnetic ordering and large volume changes in the crystal structure. Conclusively, ICT oxides like perovskites-based PCMs are promising for TES and efficient refrigeration system as the transition temperature of these compounds can be adjusted by varying the mole fraction of its chemical substitution. Further, these compounds can also be suitable for various solid-state devices with a range of operating temperatures.

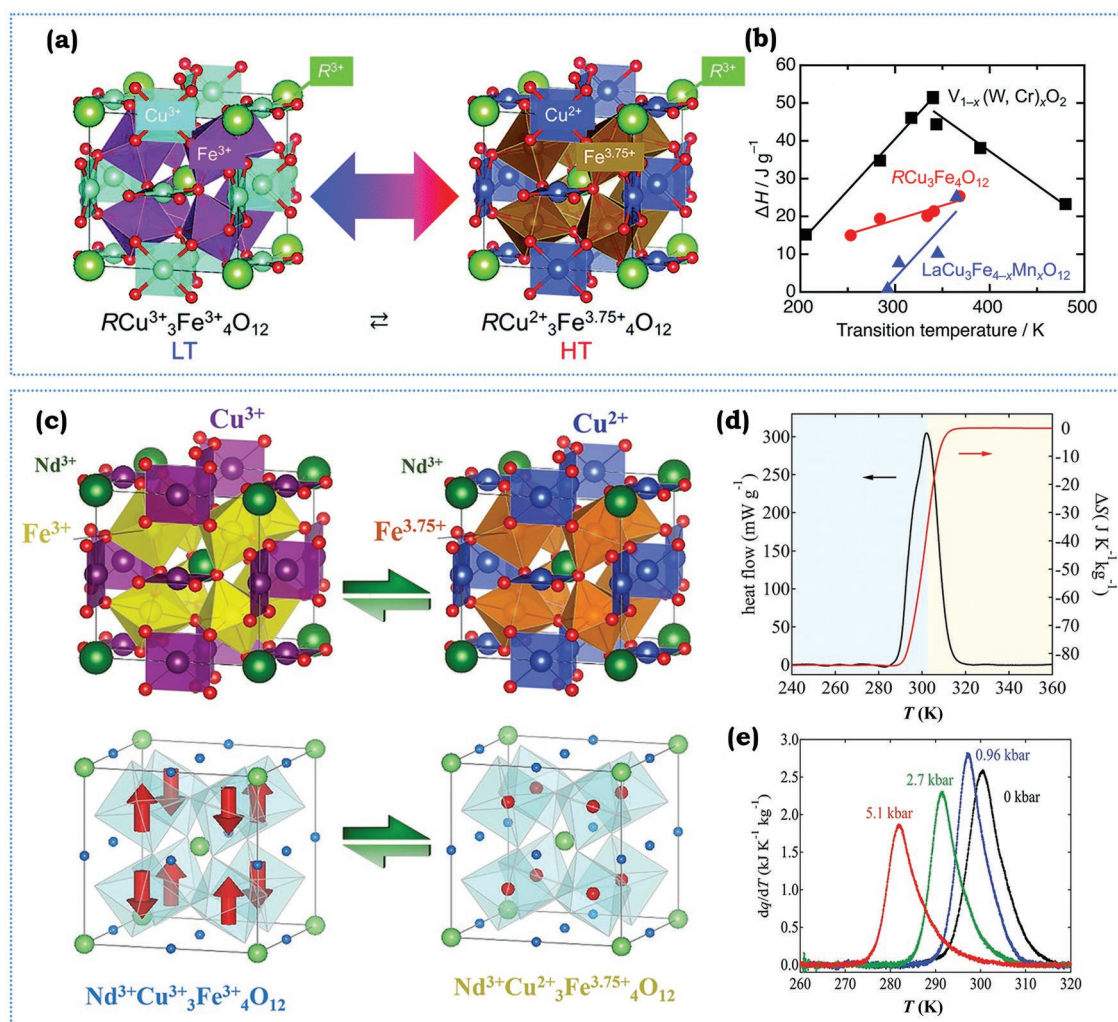


Figure 10. Organic–inorganic hybrid perovskites oxides for use in TES and refrigeration purposes through BCE effect. a) Representations of crystal structure transformation of $RCu_3Fe_4O_{12}$ between low to high temperature and vice versa through an intersite charge-transfer phase transition. b) Specific phase-change enthalpy (ΔH) over a function of transition temperature for $RCu_3Fe_4O_{12}$ and R may be La, Pr, Nd, Sm, and G indicated by the red circles, black squares indicate $V_{1-x}(W, Cr)_xO_2$, and $LaCu_3Fe_{4-x}Mn_xO_{12}$ is represented blue triangles. a,b) Adapted with permission.^[181] Copyright 2020, Royal Society of Chemistry. c) Crystal and magnetic structures of $NdCu_3Fe_4O_{12}$; the low-temperature G-type antiferromagnetic $Nd^{3+}Cu^{3+}_3Fe^{3+}_4O_{12}$ changes to the high-temperature paramagnetic $Nd^{3+}Cu^{2+}_3Fe^{3.75+}_4O_{12}$ due to the ICT transition at 37 °C showing the temperature-dependence behavior of $NdCu_3Fe_4O_{12}$. d) DSC curve of $NdCu_3Fe_4O_{12}$ at ambient temperature calculated by entropy change ΔS shown by a red line and also measured at 360 K during cooling. e) Heat flow curves divided by cooling rate in $NdCu_3Fe_4O_{12}$. c–e) Adapted with permission.^[182] Copyright 2021, Wiley-VCH.

4. Challenges Associated with S–S PCMs

The implementation of S–S PCMs in thermal-energy applications has increased significantly in recent years, owing to their inimitable advantages: solid-state processing, negligible volume change during phase transition, and long cyclic life. Despite these outstanding features, the adoption of these materials in specific engineering applications faces some issues such as lower phase-change enthalpy, and lower thermal conductivity typically in the case of polymeric- and polyalcohols-based PCMs, phase-change kinetics, ease of production, and cost in case of hybrid and inorganic S–S PCMs. Thermal conductivity is a very basic parametric means of thermo-physical properties of PCMs and it is directly linked to the heat transfer rate of the materials.^[183] Thus, thermal conductivity is one of the

key parameters that must be highlighted in the selection of S–S PCMs for TES applications.

The thermal conductivity of inorganic PCMs such as metals, ceramics, and alloys is relatively higher than organic ones, therefore, less attention has been paid to increasing the thermal conductivity of inorganic-based PCMs. On the other hand, most of the polymeric- and polyalcohols-based PCMs exhibit relatively low thermal conductivity, similar to S–L PCMs. For instance, the thermal conductivity of various polymeric S–S PCMs was found in the range of 0.10–0.15 $W m^{-1} K^{-1}$,^[43,184] which is comparable to S–L PCMs such as organic S–L PCMs, paraffin, and fatty acid. This is mainly attributed due to resemblances in counterparts or molecular structures of polymeric S–S PCMs and organic S–L PCMs. So, it is highly desirable to tune the thermal conductivity of PCMs by adding some

Table 3. A summary of thermal conductivity of various S–S PCMs.

PCMs	$K_{PCM}^a)$ [W m ⁻¹ K ⁻¹]	Specific additives	Fraction	$K_{com}^b)$ [W m ⁻¹ K ⁻¹]	Enhancement [%]	Reference
NPG	0.23	CuO nanoparticle	3 wt%	0.38	149%	[189]
PG	0.42	Sn/Bi	–	0.483	065%	[190]
PG	0.41	Al ₂ O ₃ nanoparticles	1 wt%	0.52	156%	[191]
NPG	0.23	Cu/SiO ₂	–	0.33/0.295	112%	[192]
PEG-PU _s	0.31	GO	7.5 wt%	0.478	135%	[193]
PU _s	0.31	Pitch-based graphite foam	27 wt%	10.86	4300%	[188]
Erythritol	0.72	Graphite foam	25 wt%	3.77	520%	[194]
PU-3000	0.25	CNT	5 wt%	0.49	230%	[32]
PU _s	0.3	CNT	1.2 wt%	0.44	120%	[195]
PU _s	0.3	CNT sponges	19 wt%	2.4 (axial) 0.95 (radial)	960%/380%	[31]
Magnesium-based organometallic	0.23	Nano-gallium	5 wt%	0.538	180%	[167]

^{a)} K_{PCM} represents thermal conductivity of PCMs; ^{b)} K_{PCM} represents thermal conductivity of PCMs composites.

additives or mixing two or more S–S PCMs from the same family or vice versa to improve their thermo-physical properties toward enhanced TES performance. Presently, thermal conductivity enhancement of S–S PCMs by using additives, a hybrid design model, and dynamic tuning possibilities are available to make it more viable for potential TES applications.^[185]

4.1. Thermal Conductivity Enhancement Strategies

Thermal conductivity enhancement of S–S PCMs has been typically governed by using the S–L PCMs technique and mainly two kinds of additives are being used in this regard: carbon- and metals-based geometries. There are mainly two types of strategies involved in tuning the thermal conductivity of PCMs by composition with conductive fillers: 1) additives and 2) encapsulating into porous matrixes. Regarding this, considerable efforts have been devoted to increasing the thermal conductivity of PCMs.^[183,186,187]

4.2. Carbon-Based Additives

Carbon-based additives include various demission structures. For instance, 0D refers to nanoparticles, 1D indicates nanotubes and nanowires, 2D included nanosheets, and porous 3D. These additives are usually considered an optimum choice to develop the composite with PCMs due to the advantages of high thermal conductivity, and extensive availability. For instance, a pitch-based graphite foam was effectively incorporated with PU_s-based S–S PCMs to develop a phase-change composite by our group.^[188] The thermal conductivity of the prepared composites was significantly enhanced up to 10.64 W m⁻¹ K⁻¹ which is 42 times pristine PU_s. Lately, highly conductive and flexible phase-change composites were also designed by our group by infiltrating PU as S–S PCMs into the pores of the CNT sponge. The author's results exhibited that the thermal conductivity was observed at around 2.4 in the axial direction due to the directional alignment of CNT in a polymer matrix.^[31] The summary of thermal conductivity of various S–S PCMs by using different additives is presented in **Table 3**.

4.3. Metal-Based Additives

The thermal conductivity of PCMs can also be tuned by using metals-based additives including metal foam, metal nanoparticles, and metal oxide nanoparticles. In this regard, extensive effort has been made by using metal foams and metal nanoparticles for improving the thermal conductivity of PCMs with reasonable change in enthalpy and transition temperature.^[196,197]

To conclude, carbon-based materials are relatively more significantly used for promising the composites rather than metal-based-additives, as they earned more favorable attitudes toward potential TES applications due to sealant characteristics: diversity of allotropies, good thermal stability, and high thermal conductivity. Although, metallic materials generally present higher thermal conductivity, while bearing some issues as well like high density and chemical activity that ultimately limits their practicability in various scenarios. Apart from the type of additives, some other factors may also influence the enhancement of thermal conductivity such as dispersity, functionalization, synthesis methods, and degree of extent of alignment.

5. Summary and Outlook

TES technologies play an important role in the sustainable development of modern societies. Among various methods of energy storage, phase-change energy storage is one of the actively followed technologies in terms of storage density, economic viability, and industrial maturity. Despite these outstanding features, the widespread adoption of this technology is adversely limited by volume expansion, phase segregation, and leakage problems associated with conventional S–L PCMs. Being a promising alternative to S–L PCMs, tremendous attention has been paid to developing S–S PCMs owing to their inimitable advantages: solid-state processing, negligible volume change during phase transition, and long cyclic life. Further, these PCMs also offer inimitable opportunities to develop novel and innovative materials that can syndicate energy storage applications with other multifunctional attributes for practical realizations. In this review,

we have provided a holistic understanding of the various families of S–S PCMs at the nanoscale to identify the design principle of materials for TES applications. For this, S–S PCMs are categorized into three major classes including organic, inorganic, and hybrid based on their intrinsic structural properties and topological arrangement. Further, the relationship between molecular-crystal structures with thermal properties of S–S PCMs was also critically reviewed. In summary, we have reviewed the sig-

nificant recent progress of various S–S PCMs in terms of their thermal storage applications including solid-state refrigeration technology, multifunctional devices, and recovery of exhaust heat from industrial power plants. Despite the recent development of various S–S PCMs in the applied and fundamental research field, several challenges still need to be addressed in future research. Each family of S–S PCMs has its benefits and demerits which are extensively explained (Figure 11).

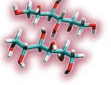
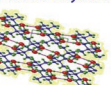
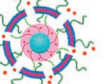
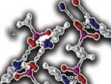
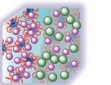
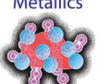
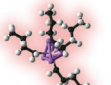
Organic S-S PCMs			
Polyalcohols  Plastic Crystal 	Benefits ✓ First-order solid-state transitions ✓ reversible change in isothermal entropy ✓ No condensations ✓ no erosion to the device ✓ Ease of production ✓ Longer life span ✓ Eco-friendly nature	Challenges ✗ Relatively low thermal conductivity ✗ Small Isothermal entropy changes ✗ Relatively High phase transition temperature	Future directions ➔ Engineering design optimization is needed to minimize the thermal hysteresis.
Polymeric S-S PCMs  	Benefits ✓ Robust mechanical properties ✓ Excellent thermal recyclability ✓ Solid-state plasticity ✓ No leakage ✓ No segregation problems ✓ Good malleable properties ✓ Superior design flexibility	Challenges ✗ Low thermal conductivity ✗ Limited energy storage capacity ✗ Relatively complex synthesis schemes ✗ Poor fire resistance	Future directions ➔ Improved thermal conductivity by tuning the intrinsic structure ➔ Developing innovative and more adaptive polymeric materials ➔ Exploring the long-term stability of these compounds at commercial scale ➔ Assembling the structure-property relationships
Organic Salt 	Benefits ✓ Chemically and thermally stable ✓ Relatively non-volatile ✓ Good to fire resistance ✓ High cycling stability ✓ Non-corrosiveness in nature	Challenges ✗ Extremely low heat storage capacity ✗ Low thermal conductivity ✗ Costly	Future directions ➔ Improved thermal storage capacity by modifying the inherent arrangements
Inorganic S-S PCMs			
Ceramics  Metallics 	Benefits ✓ Good thermo-optically responsive ✓ Good formability and ductility ✓ Instinctively remain solids during phase transitions ✓ High cycling stability and tunability ✓ Excellent erosion resistance ✓ Relatively good non-toxicity	Challenges ✗ Very high phase transition temperature ✗ Limited thermal storage capability ✗ Slightly expensive ✗ Ambiguous intrinsic structure-property relationship ✗ Insufficient studies on metallic transformation	Future directions ➔ Exploring the translucent behavior of inorganic S-S PCMs ➔ Managing the design of crystallographic materials for targeting applications
Hybrid S-S PCMs			
Organometallic  Perovskites 	Benefits ✓ Fast heat-charging dynamics ✓ High carrier mobility for thermal and mechanical stability ✓ Small changes in volumes ✓ No volatilization ✓ Relatively high enthalpies ✓ Better thermal conductivity ✓ Improved flame retardancy	Challenges ✗ Weakly connected inorganic entities with organic moieties like N-H... Cl bonds ✗ Relative complex synthetic schemes ✗ Relatively poor thermal stability ✗ Relatively high-cost	Future directions ➔ Investigating the internal intercalations between organic and inorganic counterparts for highly packing system

Figure 11. Summary of three different classes of S–S PCMs in terms of their advantages, challenges, and future goals.

In the exploration of advanced S–S PCMs, future research studies can be designed by considering the following aspects;

- 1) Apart from the large latent heat of polyalcohols-based S–S PCMs, pristine PCs and polyalcohols (like PE, NPG) that are easily sublimated after T_m which restricts their practical application to some extent in TES, can be addressed in the future. In addition, the research on the adaptability of PCs is still at an earlier stage, and further direction can be made to advance their performance at a commercial level by tuning the crystal engineering structure with special emphasis on the molecular and supramolecular factors.
- 2) There is a lack of study to rationalize the use of S–S PCMs by considering economic analysis, feasibility, and scalability from laboratory inventions to commercial end-user products. Regarding this, a comprehensive economic analysis of S–S PCMs is still missing in the literature and should be identified for global acceptance.
- 3) Dynamic cross-linked polymeric S–S PCMs are relatively more stiff, reliable, and thermally recyclable; consequently, it has strong chemical stability against the surrounding media rather than traditional PCMs. Yet, the LHS capacity of these PCMs is not at a reasonable level to some extent. Consequently, such kinds of PCMs still need further investigation.
- 4) The execution of new technology, for example, solid-state refrigeration with ease of order–disorder switching by using PCs, metal-free antiferroelectric materials, provides attractive research directions in the future for developing S–S PCMs.
- 5) As is known, a significant amount of heat is produced from an energy-related power plant or exhausting heat by industrial factories, which is being wasted without useful application. Regarding this, the use of inorganic and hybrid-based materials is integrated to make it useful for household applications, that is, water heating through the heat recovering process. However, further research is still needed to optimize this technology at a commercial level.
- 6) Inorganic S–S PCMs have been employed in smart windows for optical modulations in terms of energy-saving and energy storage owing to their thermo-stimuli-responsive behavior in molecular crystals. Nevertheless, such materials have been scarcely reported, especially for application. So, further optimization is needed to tune the intrinsic properties relationship for advanced utilization at a broader scale.
- 7) S–S PCMs refer to PCMs that can maintain a solid appearance under and over melting temperature during reversible phase transition. Lots of works about S–S PCMs have been reported focusing on their enthalpy, melting temperature, shape stability, functional modifications, integrated application, and so on, among which a scientific or quantitative description or measurement of the solid–solid phase transition property is still lacking, and is of course worth studying on.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

solid-state phase-change materials, structure–property relationship, thermal-energy applications

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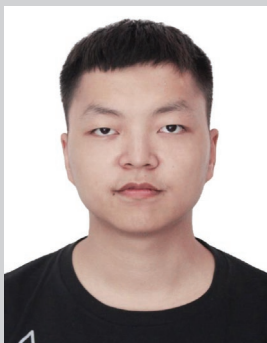
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